Gillespie-Based Simulation and Inference for Non-Markovian Stochastic Reaction Networks

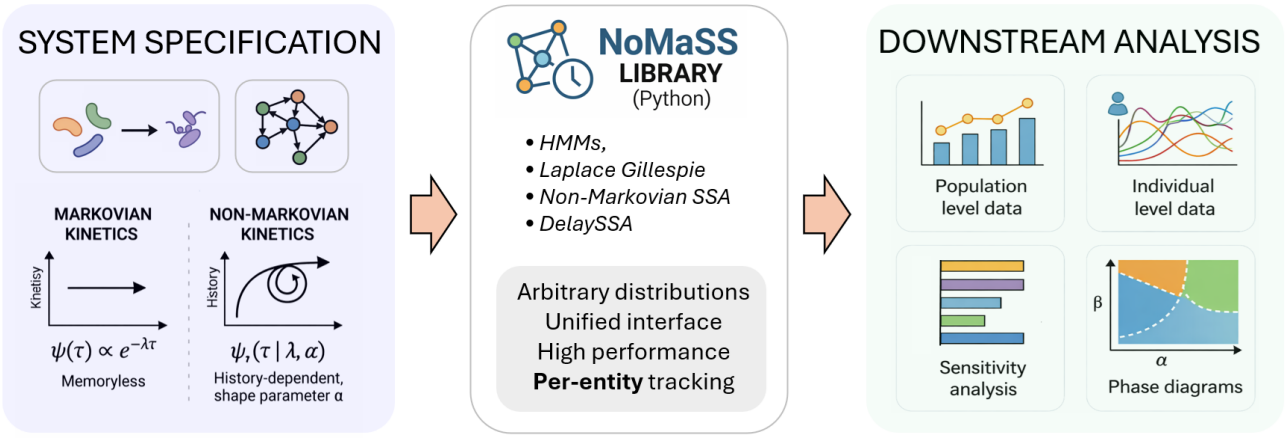
Aurélien Pélissier[1,2,3], Miroslav Phan[1,2], Niko Beerenwinkel[2] and María Rodríguez Martínez[1,4,†]

[1] *IBM Research Europe, 8803 Rüschlikon, Switzerland*
[2] *Department of Biosystems Science and Engineering, ETH Zurich, 4058 Basel, Switzerland*
[3] *Institute of Computational Life Sciences, Zürich University of Applied Sciences (ZHAW), 8820, Wädenswil, Switzerland*
[4] *Department of Biomedical Informatics & Data Science, Yale School of Medicine, New Haven, CT, United States.*
[†] *Corresponding author maria.rodriguezmartinez@yale.edu*

Abstract

Discrete stochastic processes are widespread across physics, chemistry, ecology and beyond. In computational biology and epidemiology, however, most simulators still assume Markovian kinetics with memoryless dynamics, despite growing evidence for history-dependent effects in gene regulation, RNA transcription, cell differentiation, and infection. This reliance on Markovian models limits the routine use and comparison of more realistic, memory-aware descriptions. Here, we develop and benchmark a unified framework for simulating non-Markovian reaction networks using Gillespie-based algorithms. We implement multiple algorithmic classes, including exact, rejection-based, delay-based, and hybrid Markovian/non-Markovian schemes, and compare them across representative biological and epidemiological models. Across three case studies, we show that non-Markovian waiting times can qualitatively change population-level predictions, and that delay-based approximations can break down when intrinsic system timescales approach the imposed delays. We further show how population-level measurements can be used to infer otherwise inaccessible waiting-time distributions, and how non-Markovian structure can be leveraged for sensitivity analysis and statistical inference. To support broad use of these approaches, we release `NoMaSS` (Non-Markovian Stochastic Simulations), an open-source Python library that provides a unified interface to a wide range of non-Markovian Gillespie algorithms and enables hybrid Markovian/non-Markovian models (`https://github.com/AI-SysBio/NoMaSS`).

keywords: non-Markovian stochastic simulation; Gillespie algorithm; renewal processes; delayed reactions; systems biology.



Discrete stochastic processes [1] provide a natural framework for modeling random phenomena in physics [2], biochemistry [3], epidemiology [4], finance [5] and meteorology [6]. In computational biology, they underpin widely used models of gene regulatory networks, signal transduction, and cellular decision-making. To make such systems

analytically and numerically tractable, it is common to impose the Markovian assumption, under which the dynamics are memoryless: the probability of future events depends only on the current state and not on how the system arrived there. In continuous time, this implies exponentially distributed inter-event times and therefore Poisson-like event statistics, where events occur independently and without intrinsic temporal correlations. This framework leads directly to the classical stochastic Gillespie simulation algorithm (SSA) [7, 8].

Under Markovian kinetics, the Gillespie SSA generates statistically exact realizations of reaction networks with constant reaction propensities and is typically far more efficient than fixed–time-step, agent-based simulations [9]. In its standard formulation, the system's state $\mathbf{X}(t))$ determines reaction propensities $a_r(\mathbf{X}(t))$, which in turn define a total rate a_0. At each step, the algorithm samples the next reaction time and selects the reaction to fire according to these rates, then updates the state accordingly. This Gillespie SSA has become the workhorse for stochastic modeling in systems and synthetic biology [10, 11, 12, 13, 14, 15].

However, many biological systems exhibit pronounced history dependence that violates the Markovian assumption. Hidden molecular states, multi-step biochemical reactions, chromatin remodeling, cell-cycle progression, or refractory periods in neuronal and immune responses all give rise to waiting-time distributions that deviate from the exponential law. In such systems, events are not memoryless: the probability of an event can depend on how long the system has been in its current state, leading to age-dependent or long-tailed waiting times and, consequently, non-Poissonian dynamics. Non-Markovian effects have been documented in a broad range of applications, including quantum devices [16], polymer reactions [17], molecular dynamics [18, 19, 20], biochemical reactions in single cells [21, 22], RNA transcription [23], neuronal firing [24], social interactions [25, 26], human activity patterns [27] and earthquakes [28]. In these settings, purely Markovian models can misrepresent variability, temporal correlations and even qualitative dynamical features.

While analytical approaches can capture certain classes of such non-Markovian effects, especially at steady state [29, 29], in most realistic systems, one must resort to numerical simulation. Over the past two decades, several Gillespie-type strategies have been proposed to accommodate non-exponential inter-event times. Although they share a common "one event at a time" philosophy, they differ substantially in how they encode memory. Broadly, they can be grouped into three broad classes.

Class I: Markovian embeddings with hidden or auxiliary states. The first class maintains Markovian dynamics by expanding the state space to include additional variables that account for non-Markovian effects. Hidden Markov models (HMMs) are a discrete-time example: complex temporal dependencies in the observations are captured by Markovian dynamics on a set of hidden states that act as an implicit clock or memory [30, 31]. By chaining multiple hidden states in sequence, such models generate Erlang- or gamma-distributed waiting times. Furthermore, introducing loops or branching between hidden states allows the waiting-time distribution to deviate from simple gamma forms, producing more flexible shapes determined by the transition structure.

In continuous time, a similar idea underlies approaches that represent a non-exponential waiting-time distribution as a combination of exponential phases or as a mixture of reaction rates. The Laplace–Gillespie algorithm [32] makes this explicit by modeling the density of the inter-event-time distribution (IED) as a mixture of exponentials,

$$\psi(\tau) = \int_0^\infty \lambda e^{-\lambda\tau}\, p(\lambda)\, \mathrm{d}\lambda, \qquad [1]$$

where the rates λ can be viewed as latent "tasks" or modes with different priority and $p(\lambda)$ as their distribution. At each SSA step, an effective rate is drawn from $p(\lambda)$ and a standard Gillespie update is performed with these instantaneous rates, yielding exact non-exponential inter-event times while preserving Markovian dynamics in the augmented state space and naturally capturing bursty, priority-queue–like agent behavior.

Class II: explicit timing via renewal and hazard-based methods. The second class encodes the timing of events directly, without an explicit Markovian assumption and without discretizing the state space into a series of auxiliary stages to mimic memory. Each individual process i is treated as a renewal process with a prescribed waiting-time distribution $\psi_i(\tau)$, equivalently characterized by its hazard, i.e., its age-dependent instantaneous rate.

$$\lambda_i(\tau) = \frac{\psi_i(\tau)}{1 - \int_0^\tau \psi_i(u)\, \mathrm{d}u}.$$

The non-Markovian Gillespie algorithm (nMGA) [33] and its more efficient variant MOSAIC [34] are representative examples. While the nMGA requires updating the hazards of all reaction channels at every simulation step, MOSAIC utilizes rejection sampling to eliminate these global updates. Both methods track the elapsed time τ_i since the last firing of each process, compute the current hazards $\lambda_i(\tau_i)$, and sample the next event using a Gillespie-based scheme under a short-interval approximation in which hazards are treated as locally constant. The effective step size is controlled internally: smaller time jumps reduce the error introduced by freezing the hazards, at the expense of additional computational cost. MOSAIC adaptively enforces sufficiently small steps so that the remaining approximation error is negligible in practice [34].

Because inter-event time distributions enter explicitly through their hazards, these approaches can accommodate arbitrary waiting-time laws, heterogeneous populations, and couplings to macro-variables, all without time discretization. Their main challenges are algorithmic, namely controlling the per-event computational cost and ensuring that the underlying approximations remain valid in the regimes of interest.

Class III: hard-coded delayed reactions. The third class introduces non-Markovian effects via hard-coded delayed reactions [35, 36, 37, 38, 39]. Namely, when a delayed reaction fires, its effect is not applied immediately; instead, the algorithm samples a delay from a specified distribution and enqueues a completion event (i.e., places it in a queue) to be triggered at a fixed future time. Simulation then proceeds by interleaving two mechanisms: standard Gillespie SSA sampling for instantaneous (Markovian) reactions, and the execution of a priority queue of scheduled completions.

From the perspective of the underlying stochastic model, this hybrid approach is simple, intuitive, and straightforward

to implement. However, it departs from the instantaneous-reaction character of classical Gillespie dynamics: once a delayed event is scheduled, it will occur at its assigned completion time, even if the system evolves in ways that would alter or cancel it. Delay-based schemes are therefore exact only under the assumption that the delay distribution remains fixed after initiation and that cancellations are either negligible or explicitly modeled. They also incur memory and sorting overheads when many delayed events are involved.

Bridging methods and practice. From the perspective of a typical user in bioinformatics or systems biology, the landscape of Gillespie-type extensions for non-Markovian dynamics remains fragmented. Methods are dispersed across communities, implemented in different languages, and often tailored to narrow application domains, which makes them difficult to adopt, combine, and evaluate in a consistent workflow. Existing software such as the DelaySSA toolkit [39] has lowered the barrier to using delayed reactions, but does not natively cover the broader range of Markovian-based memory approximations and hazard-based approaches within a single framework. As a result, practitioners often face a practical bottleneck: deciding which modeling formalism is appropriate for a given biological question, understanding the assumptions that come with each simulator, and integrating memoryless and history-dependent reactions within the same model.

In this article, we bridge this gap by introducing an open-source Python library, `NoMaSS`, as a computational biology and bioinformatics framework that brings multiple Gillespie-type algorithms for Markovian and non-Markovian processes under a unified interface and supports hybrid reaction networks mixing instantaneous and history-dependent kinetics. The library implements representative methods from the three classes, allows users to specify arbitrary inter-event time distributions, and provides building blocks for downstream analyses commonly needed in practice. We illustrate these capabilities on three biologically motivated case studies, i.e. microbial growth dynamics, stem cell differentiation, and RNA transcription, highlighting workflows such as sensitivity analysis and inference procedures that recover otherwise unmeasurable waiting-time distributions from population-level observations. Alongside this practical guide, we include targeted benchmarks and decision rules to help researchers select, implement, and diagnose non-Markovian simulation approaches in bioinformatics and systems biology.

NoMaSS: A Framework for Non-Markovian Stochastic Simulations

We consider a reaction network composed of reaction channels of the form

$$X \xrightarrow{r} P_1 + \cdots + P_k,$$

where X denotes one or several reactant species and $P_1, \ldots, P_k$ the products of reaction channel r. For each channel r, we model the inter-event time τ between successive reaction events using a waiting-time distribution

$$\psi_r(\tau \mid \lambda, \alpha),$$

parameterized by a nominal rate λ and a shape parameter α. The rate parameter λ sets the characteristic time scale of the reaction. For distributions with a finite mean, we impose $\mathbb{E}[\tau] = 1/\lambda$. In the exponential (Poisson) case, this corresponds to a constant hazard λ, recovering the classical Markovian memoryless assumption. For heavy-tailed distributions or distributions with undefined mean, we instead set $\text{median}(\tau) = 1/\lambda$, so that λ consistently defines a typical time scale across distribution families. The shape parameter α controls deviations from exponential kinetics by modulating the variability and tail behavior of ψ_r. Thus, compared with a standard Gillespie reaction channel, which is typically specified by a single rate parameter, the NoMaSS parametrization usually adds only one additional interpretable shape parameter per non-Markovian reaction channel. For the Gamma and Weibull families, $\alpha = 1$ recovers exponential waiting times, whereas $\alpha \neq 1$ yields more regular dynamics (narrower distributions) or burstier dynamics (broader distributions). For other families such as Normal, Lognormal, Cauchy, and power-law IEDs, α instead tunes dispersion or tail-heaviness without an exponential limit, enabling intrinsically non-exponential waiting times. Figure 1 illustrates this parametrization. Each reaction channel is driven by a rate λ_i, which sets the characteristic time scale, while additional shape parameters α_i encode non-Markovian effects.

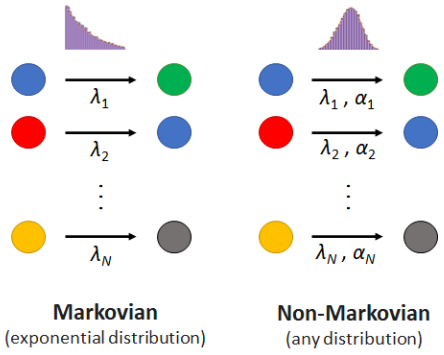


Figure 1: Schematic illustration of Markovian versus non-Markovian stochastic reaction dynamics. **Left:** In a Markovian reaction network, reactants (blue, red, yellow, ...) transform into products through N reaction channels with constant instantaneous rates λ_i, leading to exponentially distributed inter-event times. **Right:** In non-Markovian systems, inter-event times follow more complex waiting-time distributions parameterized, for example, by λ_i and additional shape parameters α_i. The resulting instantaneous rates depend on the elapsed time since the last event and, potentially, on other state variables.

`NoMaSS` is a Python library that provides a unified interface to several Gillespie-based approaches for implementing such reaction channels with arbitrary IEDs. At the modeling level, users specify a reaction network, assign a state-dependent IED to each reaction, and choose a simulation scheme for the corresponding subset of reactions. At the algorithmic level, NoMaSS provides a unified API for adding or removing reactions, switching between simulation methods, and combining Markovian and non-Markovian reactions within the same system. In practice, the NoMaSS workflow consists of four main steps: defining the reaction network, assigning an inter-event time distribution to each reaction channel, selecting the appropriate simulation backend, and using the resulting trajectories for benchmarking, parameter inference, sensitivity analysis, or biological interpretation. Furthermore, NoMaSS supports arbitrary IEDs (Table 1) and currently provides four complementary simulation back-ends:

(i) **Laplace–Gillespie.** For reactions whose waiting-time distributions admit a mixture-of-exponentials representation with a monotone decreasing hazard, NoMaSS implements the Laplace–Gillespie scheme [32]. This enables exact simulation of a broad class of long-tailed IEDs by sampling an effective rate from a mixing distribution and then performing a standard Gillespie step. In practice, NoMaSS includes built-in support for common cases such as power-laws ($\propto 1/t^{\alpha}$), as well as gamma and Weibull IEDs in the decreasing-hazard regime (e.g. gamma with $\alpha < 1$ and Weibull with $\alpha < 1$).

(ii) **MOSAIC.** For arbitrary IEDs specified via hazard functions, NoMaSS includes MOSAIC [34], an optimized variant of the non-Markovian Gillespie algorithm [33]. MOSAIC matches the SSA's asymptotic complexity and is particularly well-suited when the instantaneous rate remains finite at $\tau = 0$, such as gamma and Weibull IEDs with $\alpha \geq 1$, as well as log-normal, Pareto, and other commonly used distribution families.

(iii) **Delayed-reaction schemes.** NoMaSS implements Anderson's *modified next reaction method* [37], which encodes non-Markovianity by scheduling delayed completion events (e.g., in a priority queue) and updating the system state when those events fire. The method can also be combined with other simulation schemes in hybrid models.

(iv) **Hidden Markov embeddings.** While NoMaSS does not provide a dedicated HMM constructor, non-Markovian dynamics can be embedded into a Markovian model by manually introducing intermediate (hidden) states and reactions. This yields a sequence of exponential steps compatible with the classical SSA. In the common special case of a linear chain of identical exponential stages (Erlang waiting times),the same effective waiting-time law can be represented more compactly within MOSAIC by using a gamma IED with an integer shape parameter.

PDF name	DelaySSA	MOSAIC	Laplace
Exponential	✓	✓	✓
Gamma	✓	✓ ($\alpha \geq 1$)	✓ ($\alpha \leq 1$)
Weibull	✓	✓ ($\alpha \geq 1$)	✓ ($\alpha \leq 1$)
Gaussian / Normal	✓	✓	✗
Lognormal	✓	✓	✗
Cauchy	✓	✓	✗
Power-law (Pareto)	✓	✓	✗
Power-law (Lomax)	✓	✓	✓

Table 1: Supported inter-event time distributions (IEDs) for the NoMaSS back-end methods. Each distribution is parameterized by a rate λ, representing the inverse mean waiting time (or the inverse median when the mean is undefined), and a shape parameter α that characterizes deviations from Markovian dynamics. Constraints on α are indicated where applicable. Full IED parameterizations and implementation details are provided in Tables 2 & 3.

All methods share a common internal representation of reactions, state variables, and random number generation, making it straightforward to switch between algorithms or to simulate the same model under different assumptions about the inter-event time distributions. Users can select the simulation scheme on a per-reaction basis, enabling fine-grained control over which parts of the network are treated as Markovian, renewal-based, or delay-based.

Toy-model benchmark: unconstrained symmetric population growth. As an initial benchmark, we consider a simple pure-birth process $S \to 2S$, where each particle divides independently with a gamma-distributed IED of mean $\mu = 1/\lambda_{\text{div}}$ and shape α (Figure 2A). Analytically, the population grows exponentially as $N(t) \propto e^{\kappa t}$, where the effective growth rate κ is sensitive to the variance in division timing. Even when the mean division time μ is held constant, κ depends on the shape parameter α (Figure 2B, Supplementary A.2) according to:

$$\kappa = \frac{\alpha\left(2^{1/\alpha} - 1\right)}{\mu}. \qquad [2]$$

This relationship captures the transition between two physical extremes. In the Markovian limit ($\alpha = 1$), the memoryless nature of the exponential distribution yields a simple growth rate $\kappa = 1/\mu$. Conversely, as division times become more regular ($\alpha \to \infty$), the process becomes deterministic; the population doubles exactly every μ units of time, resulting in a slower asymptotic growth rate $\kappa = (\ln 2)/\mu$. Physically, this behavior arises because heavy-tailed distributions ($\alpha < 1$) allow a fraction of "fast-dividers" to dominate the population expansion early on, disproportionately accelerating the total growth. In contrast, more regular, "clock-like" division times ($\alpha > 1$) lack these early events, leading to a systematic reduction in the observed growth rate. This example illustrates how neglecting non-Markovian IEDs can systematically bias inferred kinetic parameters from population-level data.

Beyond the physical dynamics, the choice of IED shape parameter significantly impacts the numerical efficiency of the simulation. As shown in Figure 2C, the computational cost per event varies across the different backends implemented in NoMaSS. As expected, classical SSA is fastest because it requires neither rejection sampling nor queue management. Laplace-Gillespie and DelaySSA show comparable performance across shape parameters, reflecting their similar tree-based bookkeeping, i.e. rates in the former and scheduled completion times in the latter. In contrast, MOSAIC's runtime depends strongly on the IED through the rejection rate. For gamma IEDs, the expected number of rejections increases approximately linearly with α (Table 3), so narrower distributions lead to higher computational cost.

Overall, for IEDs that cannot be handled efficiently by the Laplace–Gillespie method, DelaySSA is typically the fastest option. This advantage, however, relies on the delay-based assumption that an event's completion time is fixed at initiation, i.e. the IED does not change after the event has been scheduled. This assumption holds for the present renewal division process but may be violated in models in which hazards depend on evolving state variables. In such cases, MOSAIC provides the more appropriate simulation scheme.

Use cases of NoMaSS in real biological systems

Application I: When delay-based approaches fail in state-dependent microbial growth. We now consider the population dynamics of the bacterium *Caulobacter crescentus* [40], which divides asymmetrically into a

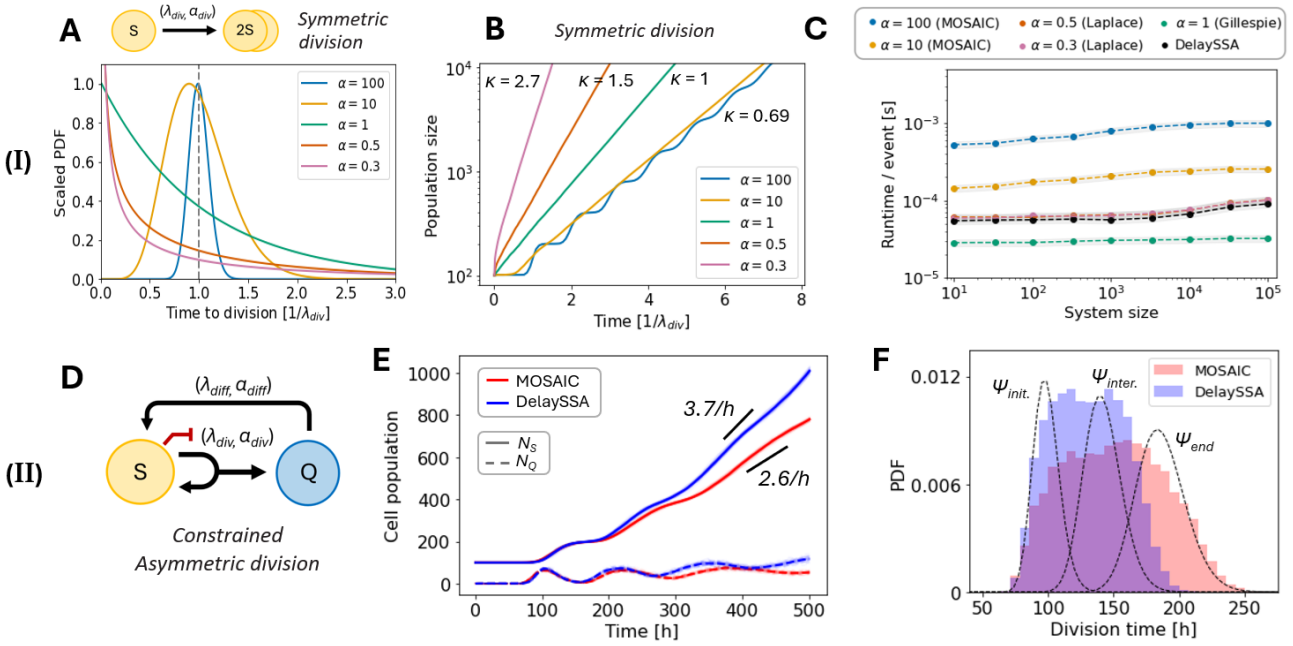


Figure 2: Dynamics of (I) symmetric and (II) asymmetric growth under non-Markovian division. (A) Top: schematic of unconstrained symmetric division ($S \to 2S$). Bottom: gamma-distributed interdivision times for different shape parameters α, with each probability density function (PDF) scaled to unit maximum for visual clarity. (B) Simulated population growth from 100 initial reactants undergoing unconstrained symmetric division ($S \to 2S$) with gamma-distributed interdivision times at fixed mean division time and varying α. Corresponding fitted growth rates (κ) are shown for reference. (C) Per-event computational cost versus system size for different methods implemented in NoMaSS for the same unconstrained symmetric division process ($S \to 2S$). Shaded bands denote the standard deviation across simulation replicates. (D) Schematic of crowd-limited microbial growth in *Caulobacter crescentus*: a stalked cell (S) divides into a new stalked cell and a motile, quiescent cell (Q), which cannot divide but can differentiate into a stalked cell. Increasing cell density reduces the division rate. (E) Population dynamics starting from 100 stalked cells ($1/\lambda_{\mathrm{div},0} = 71.6$, $1/\lambda_{\mathrm{diff}} = 28.7$, $K = 500$), averaged over 50 simulations, simulated using the MOSAIC and DelaySSA backends of NoMaSS. Asymptotic linear growth-rate fits are shown for reference. We observe a divergence in asymptotic growth rates, where DelaySSA overestimates the population growth by failing to account for increasing density during the division waiting period. (F) Density histograms of aggregated stalked-cell division-time distributions for both methods. Dashed curves represent the effective IEDs calculated by evaluating $\lambda_{\mathrm{div}}(N)$ at the population counts corresponding to the beginning, middle, and end of the simulation (Eq. 3). These curves illustrate the shift toward longer inter-division times as growth becomes constrained.

replication-competent stalked cell (S) and a motile quiescent cell (Q) that does not divide immediately but can later differentiate into a stalked cell (Figure 2D). Before a stalked cell can divide again, it must complete several prerequisite steps, including DNA replication, chromosome segregation, and physical separation [41], and therefore, division cannot occur immediately after birth. Hence, an exponential waiting-time model, which assigns the highest division probability at $t = 0$, is physiologically implausible. More generally, any model with a nonzero instantaneous division rate immediately after cell birth cannot capture the refractory period inherent to the cell cycle.

To represent the true division dynamics, we use single-cell division-time measurements for stalked cells under dilute (i.e., unconstrained, low-density) conditions [40] to calibrate the baseline inter-division-time distribution. Among the candidate distributions tested, a log-normal IED provided the best fit (Supplementary A.1, Table S1), capturing the delayed onset of division and the observed variability across individual cells. We then embed this empirically calibrated distribution into a state-dependent population growth model, in which increasing crowding slows division through a smooth density-dependent modulation of the division rate:

- **Stalked-cell asymmetric division:** $S \to S + Q$, with inter-division times drawn from a log-normal distribution. The intrinsic mean division rate $\lambda_{\mathrm{div},0}$ and shape parameter α_{div} are obtained from fits to dilute conditions [40]. To model growth limitation, we introduce a crowding factor $\phi(N) = 1 + N/K$, where $N = N_S + N_Q$ is the total population of stalked (N_S) and quiescent (N_Q) cells, and K is a carrying-capacity constant. The effective division rate is modulated as

$$\lambda_{\mathrm{div}}(N) = \frac{\lambda_{\mathrm{div},0}}{\phi(N)}, \qquad [3]$$

  ensuring that division slows as the system becomes crowded. This formulation implements *throughput-limited* dynamics: while growth is exponential at low density ($N \ll K$), the total population flux $J = N\lambda_{\mathrm{div}}(N)$ saturates at $\lambda_{\mathrm{div},0}K$ as $N \to \infty$. Consequently, the population size increases approximately linearly in the high-density regime, reflecting a constant production rate limited by available surface area or nutrients.
- **Quiescent-cell differentiation:** $Q \to S$, with non-Markovian differentiation dynamics. Because direct measurements of $Q \to S$ waiting times are experimentally challenging [42], we model differentiation using a log-normal IED with mean $1/\lambda_{\mathrm{diff}}$, which we constrain from steady-state stalked-to-quiescent population ratios [40]. For simplicity, we set the log-normal shape parameter to match that of the division process, ensuring that the relative variance in timing is consistent across all cellular transitions.

In Supplementary A.3, we show how varying the shape

parameters influences the system's growth rate and how the model's transient oscillations dampen over time, with broader IEDs reducing their amplitude. We also note that in the regime $\lambda_{\text{diff}} \gg \lambda_{\text{div}}$, the system simplifies to symmetric population growth.

This setting exposes a sharp difference between exact hazard-based simulation and delay-based approximations. Crucially, in MOSAIC non-Markovian reaction times are generated *from the current hazard* and therefore respond to changes in the system state throughout the waiting period. By contrast, in a standard DelaySSA, each division (or differentiation) event is assigned a completion time *once at initiation* and is then executed at that scheduled time, even if the environment changes in the meantime. When the rates vary slowly compared to the typical delay, this approximation is accurate; however, when the modulation is fast on the scale of the IED, DelaySSA can become strongly biased.

As illustrated in Figure 2E, our simulation reveals a regime where state-dependent growth inhibition unfolds on a timescale commensurate with the baseline cellular division time. This alignment arises because the physical drivers of crowding—the division events themselves—dictate the rate of environmental decay. Consequently, the inter-division delay is not a static parameter but a dynamic state-variable that evolves alongside the population (Figure 2F). Given a mean division time of approximately 150h within a total observation window of 500h, the system operates in a transient state where the *memory* of the environment is refreshed at a rate nearly identical to the underlying biological frequency.

In this setting, MOSAIC and DelaySSA rapidly diverge. DelaySSA systematically overestimates the number of completed divisions because many events remain "locked in" based on the favorable conditions at initiation, failing to account for the increasing system density during the waiting period (Figure 2F). This discrepancy is not merely transient: in the crowded regime, it produces an effective growth rate that significantly differs from the MOSAIC results (Figure 2E; 3.7/h for DelaySSA vs. 2.6/h for MOSAIC).

We note that MOSAIC provides a ground truth in this state-dependent context because it is a hazard-based rejection sampling scheme designed for the statistically exact realization of time-varying propensities. Its precision is limited only by the user-defined rejection bound, and we empirically confirmed numerical convergence by observing that predicted growth rates remained invariant under increasingly stringent rejection frequencies. Consequently, while the fixed-delay approximation in DelaySSA breaks down under rapid environmental modulation, MOSAIC remains robustly responsive to state-dependent feedback throughout the inter-event interval.

Overall, this example highlights a key modeling message: delay-based schemes are exact for renewal reactions with fixed post-initiation delays, but can be unreliable when inter-event statistics depend on evolving state variables (density, nutrients, stress, etc.). NoMaSS makes this distinction explicit: DelaySSA remains a convenient and often fast choice when its assumptions are satisfied, while MOSAIC provides a near exact hazard-based alternative when reaction rates must remain responsive to a changing environment.

Application II: Inference of unmeasurable differentiation times from stem cell population data.

Another interesting application for NoMaSS is stem cell differentiation, the process by which specialized cells are formed from stem cells [43]. Stochastic fluctuations in key transcription factors play important roles in the early stages of differentiation [44], but it is less clear whether progression along a differentiation trajectory is itself intrinsically stochastic or reflects a tightly orchestrated sequence of intermediate meta-stable states. The latter hypothesis could explain the apparently precise regulation of differentiation at the single-cell level.

From a modeling point of view, transitions between functional cell types can be described as a stochastic process with *memory*, where "memory" reflects how long a cell has resided in its current state before progressing. In practice, however, waiting-time distributions for individual cells are rarely observable: what is typically measured are population fractions, i.e. the proportions of cells in each sub-population at a set of time points. Here, we use NoMaSS to ask whether a previously proposed hidden Markov model (HMM) for stem cell differentiation is consistent with the data [22], and to infer the underlying single-cell waiting-time distributions directly from population dynamics.

Concretely, we consider the stochastic differentiation of mouse embryonic stem cells (ESC) along the neural lineage. The experimental data used for this application consist of the population fractions of three functional cell stages, embryonic stem cells (ESC), epiblast-like (EPI) states, and neuroprogenitor cells (NPC), measured at seven time points over a duration of approximately 200 hours [22]. The original study modeled these population dynamics using a hidden Markov model (HMM) [30], in which each observable cell type (ESC, EPI, NPC) is associated with multiple hidden microstates. Fitting the experimental data required 19 or 20 such microstates, depending on the cell line. Although this formulation is flexible, it introduces a large number of parameters (roughly one per microstate), and different combinations of transition and emission rates can lead to similar fits. Importantly, under the simplifying assumption that all transitions between microstates occur at the same rate, the HMM reduces to an Erlang (gamma) waiting-time distribution for each functional transition. In other words, the HMM encodes an implicit hypothesis about the single-cell inter-event time distribution. With NoMaSS, we can test this hypothesis directly while working with a much simpler model.

In NoMaSS, we represent the differentiation process using only two stochastic reactions (Figure 3):

- **ESC** $\rightarrow$ **EPI**, with inter-event times drawn from a distribution $\psi_1(\tau \mid \lambda_1, \alpha_1)$,
- **EPI** $\rightarrow$ **NPC**, with inter-event times drawn from $\psi_2(\tau \mid \lambda_2, \alpha_2)$.

Because these inter-event time distributions are assumed to depend only on the current cell type and not on global population variables, a delay-based scheme is appropriate: in NoMaSS we therefore use a DelaySSA back-end, in which each differentiation event is scheduled at commitment and completed after a random delay drawn from ψ_r. This directly matches the biological assumption that once a cell has committed to differentiate, its completion time is not influenced by subsequent changes in the rest of the population.

We first compare a Markovian model to a non-Markovian gamma model. In the Markovian case, both transitions are exponential, i.e. we fix $\alpha_1 = \alpha_2 = 1$ and infer rates λ_1, λ_2. In the gamma case, we infer (λ_1, α_1) and (λ_2, α_2) from the population fractions of ESC, EPI and NPC over time by min-

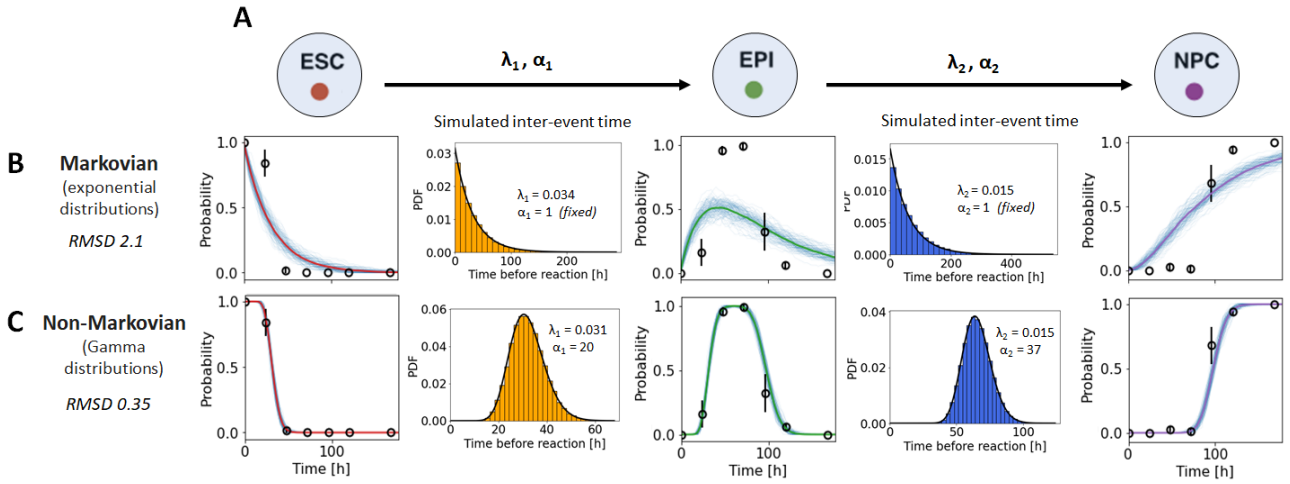


Figure 3: (A) Mouse stem cells at three stages of differentiation. From the ESC stage, cells may differentiate into EPI, which in turn may differentiate into NPC. The inter-event times associated with these two differentiation steps are modeled using either (B) exponential or (C) gamma distributions. The average root mean square deviations (RMSDs) between simulated and measured data at each time point and for each cell type are indicated for both models. The Markovian case is fitted by fixing the shape parameters $\alpha_1 = \alpha_2 = 1$ (exponential inter-event times). Black circles and error bars represent the mean and standard deviation of the stem cell populations measured at different time points [22]; solid lines show population dynamics obtained by averaging 1000 NoMaSS simulations with parameters that minimize the RMSD. Individual simulations are shown as narrow blue lines. The inferred inter-event time distributions after parameter optimization are plotted in orange and blue for the ESC→EPI and EPI→NPC transitions, respectively. Here, the exponential case peak at $t = 0$, failing to capture the biological lag or refractory period represented by the gamma distributions.

imizing the root mean square deviation (RMSD) between data and simulations. Averaging 1000 NoMaSS trajectories for each parameter set, we obtain an RMSD of 2.1 for the exponential (Markovian) model and 0.35 for the gamma (non-Markovian) model, i.e. an order-of-magnitude improvement. This difference is clearly visible in Figure 3, where only the gamma-based Non-Markovian model reproduces the timing and magnitude of the observed population changes.

We then repeat the analysis using other waiting-time families for ψ_1 and ψ_2, including normal, log-normal and Weibull distributions (Supplementary B). Briefly, the Weibull model performs substantially worse (RMSD = 0.69) than gamma (0.35), normal (0.38) and log-normal (0.39) models, which all give similarly good fits. This discrepancy arises from the Weibull distribution's polynomial increasing hazard rate; its characteristic "short tail" fails to capture the sustained stochasticity of the system and over-penalizes longer waiting times, which does not accurately reflect the underlying biological dynamics. We also examined the impact of preprocessing by fitting both the raw proportion data and a logit-transformed version (to spread points close to 0 and 1). For logit-transformed data, the gamma distribution yields the lowest RMSD; for non-transformed data, the log-normal distribution is marginally better. In both cases, however, gamma, normal, and log-normal distributions capture the data comparably well, consistent with their similar shapes in the parameter regime here inferred (Figure S1).

Finally, a gamma distribution with integer shape parameter k is equivalent to the sum of k exponential waiting times (an Erlang distribution). Such a distribution naturally arises when a process progresses through a sequence of hidden intermediate states. The good fit of an Erlang-type waiting-time model is therefore consistent with earlier HMM-based interpretations of stem cell differentiation as a multi-stage process [22]. Using NoMaSS, we can represent this behavior directly with a compact non-Markovian model and compare it with alternative waiting-time assumptions.

Application III: Quantifying the impact of non-Markovian dynamics on observed mRNA counts.

We next model transcription, the process by which DNA is copied into messenger RNA (mRNA). After RNA polymerase binds to the promoter region, a nascent RNA molecule is elongated until the full gene has been transcribed; the resulting mRNA can then be translated into protein and is eventually degraded at an approximately constant rate. Real-time measurements in single cells have revealed pronounced stochasticity in transcription [45], motivating a large body of stochastic models for gene expression [46, 23, 11].

A widely used description is the two-state or telegraph model, in which the promoter switches between an ON and an OFF state at constant rates. Transcription can occur only in the ON state, leading to bursty, non-Poissonian mRNA fluctuations that match many experimental observations [48]. This model has been studied extensively [49] and used to infer kinetic parameters for thousands of genes from single-cell RNA-seq data [50, 51]. Nevertheless, there is clear evidence that the statistics of molecular fluctuations often deviate from simple Markovian predictions [23, 52], implying that non-exponential waiting times, and thus non-Markovian descriptions, are required. This is supported by the discovery of biological *refractory periods* and non-Poissonian noise profiles that cannot be captured by constant transition rates.

NoMaSS enables us to probe directly how non-Markovian features of the underlying molecular steps influence observable mRNA distributions. To do so, we model RNA transcription using four reactions (Figure 4A):

- **Promoter activation and deactivation:** $G_{off} \xrightarrow{\lambda_{on},\, \alpha_{on}} G_{on}$ and $G_{on} \xrightarrow{\lambda_{off},\, \alpha_{off}} G_{off}$. The promoter switches between OFF and ON, with inter-event times following gamma distributions with parameters $(\lambda_{on}, \alpha_{on})$ and $(\lambda_{off}, \alpha_{off})$, respectively. These non-exponential kinetics account for the fact that promoter switching typically involves multiple sequential molecular steps, such as

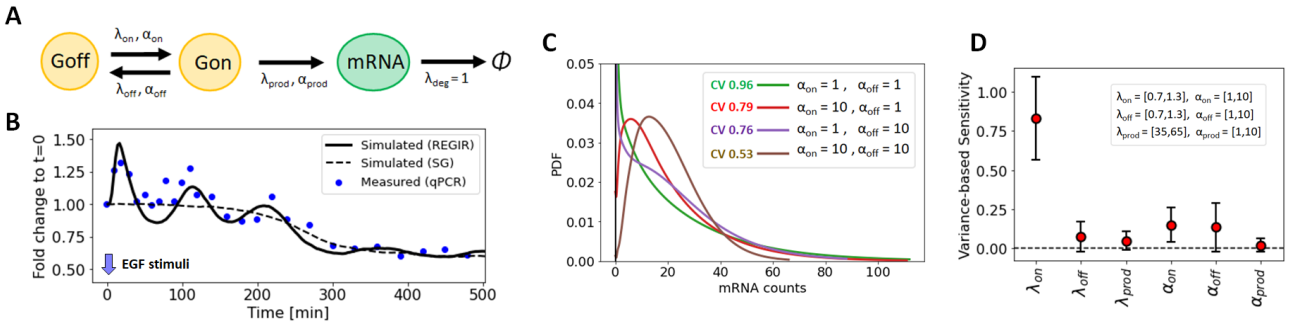


Figure 4: (A) Cartoon of the two-state model of RNA transcription with non-Markovian dynamics. The gene promoter switches between ON and OFF; mRNA is produced only in the ON state and degrades at a constant rate. (B) Average mRNA profile of the gene SLC7A5 after stimulation by epidermal growth factor (EGF) [47], measured by quantitative PCR (qPCR) and normalised to 1 at the time of stimulation. Simulated populations are averaged over 10k trajectories for both the Markovian model (SSA, RMSD = 0.016) and the non-Markovian model (NoMaSS, RMSD = 0.0074), using the best-fit parameters for each. (C) Steady-state mRNA distributions for different shape-parameter combinations, with $\lambda_{\text{on}} = 1$, $\lambda_{\text{off}} = 3$, $\lambda_{\text{prod}} = 100$ and $\alpha_{\text{prod}} = 1$ fixed. Densities are estimated by Gaussian kernel density estimation after a log transform (see Methods); the coefficient of variation (CV) is reported for each case. (D) Sobol variance-based sensitivity analysis of the CV of the steady-state mRNA distribution generated by the two-state model. The grid comprises 128 points per parameter. The rate λ_{on}, which controls promoter activation, has the strongest influence on the variance.

chromatin remodeling or transcription factor assembly, whose convolution yields gamma-like statistics.

- **RNA production:** $G_{\text{on}} \xrightarrow{\lambda_{\text{prod}},\, \alpha_{\text{prod}}} G_{\text{on}} + \text{mRNA}$. When the promoter is ON, mRNA production (initiation and elongation) is modelled by a gamma waiting-time distribution with parameters $(\lambda_{\text{prod}}, \alpha_{\text{prod}})$. This represents the cumulative time required for initiation, promoter escape, and elongation—a multi-step process that often exhibits a significant lag compared to memoryless production.
- **RNA degradation:** $\text{mRNA} \xrightarrow{\lambda_{\text{deg}}} \varnothing$. In contrast, mRNA decay is modeled with exponential waiting times. As a single-particle, first-order process where degradation is equally likely at any time regardless of the molecule's age, it is well-described by memoryless kinetics [53]

Without loss of generality, all rates are expressed in units of λ_{deg} (i.e. $\lambda_{\text{deg}} = 1$), following standard practice [50, 49].

In this application, a delay-based implementation is appropriate because we do not explicitly model energy-limited elongation or other time-dependent mechanisms that render elongation rates strongly state- or history-dependent. Prior work has identified regimes in which such energy-limited effects necessitate fully renewal/hazard-based simulation (e.g., MOSAIC) [34, 33]; those conditions are not required here.

To highlight the role of non-Markovian dynamics, we use time-resolved mRNA profiles measured by qPCR after EGF stimulation [47]. Several genes (e.g. CREG1, DDB2, SLC7A5, NFYC, NR4A1, PTGS2) display oscillatory responses that cannot be reproduced by the Markovian two-state model (Figure 4B and SI Section 5B). In contrast, after fitting the gamma shape parameters $(\alpha_{\text{on}}, \alpha_{\text{off}}, \alpha_{\text{prod}})$ with NoMaSS, the non-Markovian model can reproduce both the period and the decaying amplitude of these oscillations. The oscillation period T is largely determined by the mean switching times, $T \approx 1/\lambda_{\text{on}} + 1/\lambda_{\text{off}}$, while the damping of the oscillations is controlled by the shape parameters α_{on} and α_{off} (SI Section 5A). This pattern, seen across multiple genes [47], supports the view that promoter switching is not well described by purely Markovian kinetics [54].

Beyond fitting specific time courses, NoMaSS makes it straightforward to explore how non-Markovian features shape the steady-state distribution of mRNA counts. In the bursty regime, roughly, $\lambda_{\text{off}} > \lambda_{\text{on}}$ and $\lambda_{\text{prod}} \gg \lambda_{\text{on}}, \lambda_{\text{off}}$ [23], the shape parameters α_{on} and α_{off} have a strong impact on the distribution. For example, in Figure 4C the Markovian case ($\alpha_{\text{on}} = \alpha_{\text{off}} = 1$) yields a high probability of observing zero mRNAs, whereas increasing both shape parameters to 10 makes zero counts highly unlikely, even though the mean rates are unchanged. This illustrates the value of a framework that enables the systematic exploration of both Markovian and non-Markovian regimes within a single unified modeling environment.

To systematically quantify the parameters primarily driving variability in mRNA counts, we performed a Sobol variance-based global sensitivity analysis [55]. Unlike local sensitivity methods that evaluate effects at a single point in the parameter space, Sobol analysis decomposes the total output variance into contributions from individual parameters and their nonlinear interactions. This approach provides a comprehensive mapping of how parametric uncertainty propagates to the system's output. Because the steady-state mean of the mRNA distribution depends solely on mean waiting times and remains independent of the shape parameters [56] (Figure S8), we utilized the coefficient of variation ($CV = \sigma/\mu$) as our primary output metric (Figure 4D). Our analysis reveals that the CV is relatively robust to parameters governing mRNA production (λ_{prod}, α_{prod}). This aligns with prior analytical work demonstrating that the distribution shape is insensitive to production variations within the regime $\lambda_{\text{prod}} \gg \lambda_{\text{on}}, \lambda_{\text{off}}$ [49, 56]. In contrast, the system exhibits high sensitivity to switching kinetics: λ_{on} exerts a significantly stronger influence on the CV than λ_{off}, whereas the shape parameters α_{on} and α_{off} contribute with comparable magnitudes. These trends are further supported by phase diagrams (Figure S9), which illustrate how varying shape parameters affect the CV landscape.

Together, these results demonstrate how NoMaSS can be used to (i) fit non-Markovian transcription models to time-course data, (ii) quantify the effect of non-exponential waiting times on steady-state mRNA distributions, and (iii) perform phase-diagram and sensitivity analyses across wide parameter ranges. This is particularly important in gene

expression, where kinetic parameters span orders of magnitude [50] and non-Markovian effects can significantly distort observable mRNA statistics.

Discussion

Non-Markovian effects are increasingly recognized as a key feature of biological stochastic dynamics, yet they remain underused in everyday modeling because they substantially complicate simulation, calibration, and model comparison. Here we take a deliberately practical approach: we organize major Gillespie-type methods for non-Markovian dynamics into a small set of conceptual families, benchmark their assumptions and performance on representative biological problems, and make them accessible through a unified software interface (NoMaSS) that supports hybrid Markovian/non-Markovian systems.

A central design choice is the parameterization of waiting-time distributions as $\psi_r(\tau \mid \lambda, \alpha)$, where λ governs the mean timescale and α captures deviations from exponential, memoryless kinetics, such as increased regularity or burstiness. Across all applications, changing α while keeping the mean waiting time $\mathbb{E}[\tau]$ fixed led to substantial shifts in biologically observables, including growth rates, oscillation amplitudes, and steady-state variability.

This added flexibility should nevertheless be carefully assessed. In noisy or sparse biological datasets, non-Markovian waiting-time distributions may improve the fit by replicating technical noise, hidden heterogeneity, or outliers rather than true biological memory. We therefore recommend using the Markovian Gillespie model as the baseline and retaining non-Markovian structure only when the improvements are robust across model comparisons, held-out data, alternative waiting-time families, or sensitivity analyses. Because the Markovian case is recovered for common distributions such as Gamma and Weibull when $\alpha = 1$, this comparison is straightforward within the NoMaSS framework. NoMaSS provides a high-performance implementation that serves as a foundation for extensive downstream analyses. Its efficiency facilitates external computational pipelines—such as sensitivity analysis, e.g. Sobol indices, and phase-diagram across wide parameter ranges—that require high-throughput ensemble simulation. Simultaneously, NoMaSS supports mechanistic, event-driven studies by tracking per-entity ages and gathering individual-level statistics, providing the granular data necessary for rigorous parameter inference.

Our three case studies illustrate complementary ways in which memory shapes biological inference and prediction. In microbial growth, realistic division and differentiation time distributions changed both the effective growth rate and the degree of transient synchronization compared with mean-matched Markovian models. This example also clarifies when delay-based implementations fail: if propensities evolve on timescales comparable to the inter-event times, scheduled delays no longer reflect the current system state. In stem cell differentiation, we showed that a high-dimensional hidden Markov model (HMM) can be collapsed into a compact non-Markovian description by replacing latent progression with a gamma/Erlang inter-event distribution while still reproducing population-level dynamics. Because NoMaSS makes the Erlang–HMM correspondence explicit, comparisons with alternative waiting-time assumptions (normal, log-normal, Weibull) become transparent: differences in fit reflect distributional shape rather than differing numbers of latent states. In RNA transcription, non-exponential promoter switching altered both transient responses and steady-state mRNA distributions. Combining global sensitivity analysis with phase diagrams further revealed which kinetic rates and shape parameters control these effects, highlighting how memory in promoter dynamics propagates to observable mRNA variability.

A consistent takeaway is that delay-based methods provide a simple and often effective default for many biological systems [39], while retaining flexibility in the choice of IED. When delays are specified at reaction initiation and remain invariant to later state changes, delay schemes integrate naturally with the standard stochastic simulation algorithm (SSA). They are typically more efficient than non-Markovian Gillespie variants because simulation reduces to standard event selection plus a delay queue, avoiding rejection or resampling to handle competing non-exponential waiting times. This makes them well-suited for common systems-biology motifs such as maturation, transport, and processing steps.

Delay formulations are not universal. When propensities or waiting-time laws depend strongly on the evolving state, so that "commitment" at initiation becomes outdated, or when delays are frequently interrupted, canceled, or rescheduled, both their conceptual validity and computational advantages can break down [34]. In such regimes, instantaneous renewal formulations that continuously update hazards based on current age and state can provide a more faithful representation and may even reduce bookkeeping overhead [34, 13, 12].

Among approaches that represent non-Markovian dynamics through equivalent Markovian formulations, the Laplace–Gillespie method occupies a more specialized niche, in which waiting-time distributions are represented as mixtures of exponentials. Many biological mechanisms that generate non-Markovian dynamics, such as maturation or refractory effects, suppress very short waiting times. In contrast, the Laplace–Gillespie construction place substantial probability mass near zero [32]. This mismatch makes it less well suited for several biochemical and cellular examples considered here. Nevertheless, it remains valuable in domains characterized by strong heterogeneity in effective rates, such as earthquakes [57], epidemics [58], and bursty human interaction processes [59, 60].

From a computational perspective, all event-driven Gillespie variants can become slow for very large reaction networks or under extreme time-scale separation, where rare events are selected infrequently and steady state is reached only after long runtimes. In NoMaSS, this limitation follows from the design: because we track individual ages or initiation times and accumulate per-reactant statistics, runtime increasingly becomes dominated by bookkeeping and propensity evaluation as system size grows. Consequently, NoMaSS is not intended as a high-throughput engine for massive particle pools. Instead, it is optimized for settings where individual-level timing information is biologically meaningful and where non-Markovian effects are expected to matter.

In genuinely large-scale regimes, coarse-grained abstractions are often preferable [61, 62]. When many reactants are statistically similar, it is often unnecessary to keep track of every individual waiting time. Instead, reactants can be grouped into *bins* based on their internal *age* (i.e. the time elapsed since a reactant's last event), with hazards and event counts updated at the bin level. Such grouped-clock repre-

sentations reduce memory and scheduling overhead while preserving the key distributional effects of non-exponential waiting times. They also reflect a broader statistical principle: as group sizes increase, individual stochastic fluctuations average out and the dynamics approach a continuous (concentration-like) description, whereas fluctuations in small or critical subpopulations can dominate system-level behavior. In this large-scale limit, hybrid approximations become a natural extension, particularly piecewise deterministic Markov process (PDMP)–style schemes that treat large, fast components through deterministic continuous dynamics while retaining discrete stochasticity for rare events or small-count species [63, 64, 65]. Similar hybrid modeling strategies are implemented in platforms such as COPASI, which supports deterministic, stochastic, and hybrid deterministic–stochastic simulations of biochemical networks [66].

Finally, non-Markovian settings involving interactions between multiple reactants, most notably processes on temporal networks [67], remain comparatively less developed. A central difficulty is that the notion of an inter-event time is not uniquely defined. Waiting times may be assigned to edges (pairwise contact renewal), nodes (activity renewal), or higher-order interaction motifs, and each choice leads to a different hazard structure and corresponding simulation algorithm [59, 34]. This ambiguity helps explain why much of the temporal-network literature has focused on activity-driven models, which define node-level activation processes and generate contacts through pairing rules rather than modeling interaction events directly as competing hazards in continuous time [68]. Only recently have Gillespie-style methods been extended to accommodate non-Markovian contact processes on networks [59, 34, 69]. Systematic connections between activity-driven models and hazard-based event simulation, particularly for epidemiological dynamics on time-varying contact networks, therefore remain an open and practically important direction.

Overall, our results point to a clear modeling strategy. Begin with delay-based formulations when biological processes involve well-defined delays that remain stable after initiation. When reaction timings depend on the evolving state of the system or when events can be interrupted or rescheduled, hazard-based renewal methods provide a more faithful description. Finally, when non-exponential waiting times arise from sequences of hidden intermediate steps, they can often be represented through Markovian embeddings, allowing simulation with standard Gillespie dynamics. By placing these alternatives behind a common interface, NoMaSS enables systematic comparison of modeling assumptions and elevates memory to a testable dimension of stochastic models that can be evaluated, reported, and reproduced alongside structural and parametric assumptions.

Materials and Methods

Code and data availability. The NoMaSS implementation, input data, and scripts required to reproduce the analyses and figures presented in this article are publicly available on GitHub at `https://github.com/AI-SysBio/NoMaSS`.

Parametrization of IEDs implemented in NoMaSS `NoMaSS` implements a set of commonly used IEDs to model non-Markovian reaction dynamics, including Weibull, Gamma, Normal, Lognormal, Cauchy, and two power-law families (Pareto I and Lomax/Pareto II). For consistency across models and with prior work in stochastic simulation, each IED is specified by two parameters: a nominal rate λ and a dimensionless shape parameter α (Table 2).

The nominal rate λ sets the characteristic time scale. When the mean inter-event time is finite, we choose parameters such that $\mathbb{E}[\tau] = 1/\lambda$. For heavy-tailed distributions whose mean may be undefined (Cauchy and the two power-law cases), we instead fix the time scale using the median, $\text{median}(\tau) = 1/\lambda$. In addition to setting the typical time scale, λ also coincides with the *mean observed instantaneous rate* of the renewal process: if one samples the system at a uniformly random time, the average observed rate over this observation distribution follows $\mathbb{E}_{\text{obs}}[\lambda(\tau)] = 1/\mathbb{E}[\tau] = \lambda$ whenever the mean exists [34].

The shape parameter α controls the deviation from memoryless (exponential) kinetics by tuning dispersion and/or tail behavior, and is defined to be scale-free across all families. For Cauchy, Normal, and Lognormal IEDs we set α to the coefficient of variation, $\alpha = \text{std}(\tau)/\mathbb{E}[\tau]$, so that α directly measures relative variability. For Gamma and the power-law families, α coincides with the standard shape/tail exponent used in the usual parametrization.

MOSAIC and Laplace Gillespie implementation in NoMaSS Table 3 summarizes the operational quantities required by the simulation backends implemented in `NoMaSS`, i.e. the MOSAIC renewal scheduler [34] and the Laplace–Gillespie construction [32]. Both approaches rely on the same age-dependent propensity, given by the instantaneous rate (hazard) $\lambda(\tau) = \psi(\tau)/S(\tau)$, but use it in different ways.

The Laplace–Gillespie backend applies when the waiting-time law admits an exponential-mixture representation. In that case one can introduce a latent rate ℓ drawn from $p(\ell)$ and simulate conditionally Markovian dynamics between renewal updates, which can yield substantial speedups when many processes share the same mixture structure.

In MOSAIC, hazards are evaluated at the current "age" of each pending event and used within a rejection/thinning procedure; efficiency is controlled by an upper bound $\lambda_{\max}$ on the hazard. For IED families with bounded hazards, `NoMaSS` uses the corresponding finite $\lambda_{\max}$ directly. For families whose hazards can grow without bound, `NoMaSS` tracks the largest current age among pending processes ($\tau_{\max}$). The operative bound is then taken as $\lambda(\tau_{\max})$, so the rate bound automatically tracks the oldest active renewal clock in the system. In addition, `NoMaSS` enforces the MOSAIC accuracy criterion

$$N_r\,\lambda_{\max} \;\geq\; 20\,\lambda_r \qquad \text{for all channels } r,$$

where N_r is the number of currently eligible renewals/reactants in channel r and λ_r is the nominal rate of that channel. This requirement makes the thinning bound sufficiently conservative, effectively capping the MOSAIC step size and rendering the approximation error negligible in practice, as recommended in [34].

Earth mover distance. Also known as the Wasserstein or Kantorovich metric [72], the Earth mover distance (EMD) [73] is a measure of the distance between two probability distributions $P = [p_i]$ and $Q = [q_j]$ (i, j: index of instances in the probability distribution) over a metric space D. Where $D = [d_{i,j}]$ is the ground distance between clusters

IED name	**PDF Parametrization (nominal rate λ & shape parameter α)**
Exponential	$\psi(\tau) = \lambda e^{-\lambda\tau}$
Weibull	$\beta = \alpha\left(\lambda\,\Gamma(\frac{\alpha+1}{\alpha})\right)^{\alpha}, \quad \psi(\tau) = \beta\,\tau^{\alpha-1}\exp\left(-\frac{\beta}{\alpha}\tau^{\alpha}\right)$
Gamma	$\beta = \alpha\lambda, \quad \psi(\tau) = \frac{\beta^{\alpha}}{\Gamma(\alpha)}\,\tau^{\alpha-1}e^{-\beta\tau}$
Gaussian / Normal	$\mu = \lambda^{-1},\ \sigma = \alpha\mu, \quad \psi(\tau) = \frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\frac{(\tau-\mu)^2}{2\sigma^2}\right)$
Lognormal	$\mu = \log\left(\frac{1}{\lambda\sqrt{1+\alpha^2}}\right),\ \sigma = \sqrt{\log(1+\alpha^2)}, \quad \psi(\tau) = \frac{1}{\tau\sigma\sqrt{2\pi}}\exp\left(-\frac{(\ln\tau-\mu)^2}{2\sigma^2}\right)$
Cauchy	$\mu = \lambda^{-1},\ \gamma = \alpha\mu, \quad \psi(\tau) = \frac{1}{\pi\gamma\left(1+\left(\frac{\tau-\mu}{\gamma}\right)^2\right)}$
Power-law (Pareto I)	$\tau_0 = \lambda^{-1}2^{-1/\alpha}, \quad \psi(\tau) = \begin{cases}\alpha\,\tau_0^{\alpha}\,\tau^{-(\alpha+1)}, & \tau \geq \tau_0,\\ 0, & \tau < \tau_0\,.\end{cases}$
Power-law (Lomax / Pareto II)	$\kappa = (2^{1/\alpha}-1)\lambda, \quad \psi(\tau) = \alpha\kappa\,(1+\kappa\tau)^{-(\alpha+1)}$

Table 2: Parametrization of inter-event time distributions (IEDs) implemented in NoMaSS. Each PDF is specified by a nominal rate λ and a shape parameter α. For distributions with a finite mean, λ is chosen such that $\mathbb{E}[\tau] = 1/\lambda$. For heavy-tailed distributions without a well-defined mean (Cauchy and the two power-law cases), λ is set via the median, i.e., median$(\tau) = 1/\lambda$. The shape parameter α is dimensionless (scale-free): for Normal, Lognormal, and Cauchy it equals the coefficient of variation ($\alpha = \sigma/\mathbb{E}[\tau]$), while for Gamma, Weibull and the power-law families it matches the standard shape/tail exponent.

$[p_i]$ and $[q_j]$. EMD reflects the minimal amount of *work* that must be used to transform one distribution into the other by moving *distribution mass* around. To compute EMD, we first need to find the optimal flow $F = [f_{i,j}]$ to minimize the overall cost when moving from P to Q.

$$\min \sum_{i=1}^{n}\sum_{j=1}^{m} f_{i,j}d_{i,j} \qquad [4]$$

Finding the optimal flow F is typically related to an instance of the transportation problem, and can be efficiently solved in polynomial time [74]. EMD is then defined as the minimal work normalized by the total flow:

$$\mathrm{EMD}(P,Q) = \frac{\sum_{i=1}^{n}\sum_{j=1}^{m} f_{i,j}d_{i,j}}{\sum_{i=1}^{n}\sum_{j=1}^{m} f_{i,j}} \qquad [5]$$

Note that EMD satisfies the metric properties, i.e., positivity, symmetricity, triangle inequality, and free from any choice of bin size if $d_{\mathrm{i,j}}$ follows the metric properties

In our application, the EMD between IEDs is quantified in unit of time. However, to ensure consistency of the EMD value over different parameter space, we normalize the distance matrix $[d_{i,j}]$ by the mean inter event time $(1/\lambda_0)$ of the reaction channel. We use the Fast EMD python implementation [75] to efficiently compute the distances. The distances are measured by sampling 1×10^4 points from both the theoretical and simulated inter-event distributions, which result in an intrinsic EMD on the order of 1% (Figure S10). This comes from the fact that sampled points from equivalent distributions will not be strictly equal, which will result in a non-zero EMD.

Parameter optimization. For each NoMaSS simulation, we quantify its *error* by comparing the simulated population dynamics to the experimentally measured populations. We define the *score* of the model as the root mean square deviations (RMSD) between the measured and simulated population averaged over all timepoints and reactants. In the case of the stem cell differentiation, the populations describe proportions bounded between 0 and 1. This may bias the the optimization towards ignoring variations close to very low (~ 0) and very high (~ 1) values, as a proportion of 98 % will be considered closer to 99 % than 50 % is to 52 %. To correct for this bias, we transform both simulated and measured populations with an inverse sigmoid function. For experimental reasons, proportions lower than 1% or higher than 99% are considered not reliable enough, so we only apply the transformation within that range:

$$\begin{aligned} y_{\mathrm{new}} &= \log\left(\frac{y}{1-y}\right) \ \text{ for } \ 1\,\% \leq y \leq 99\,\% \\ y_{\mathrm{new}} &= 4.6 \ \text{ for } \ y > 99\,\% \\ y_{\mathrm{new}} &= -4.6 \ \text{ for } \ y < 1\,\% \end{aligned} \qquad [6]$$

Then, the score function is minimized with the `maxLIPO` algorithm [76], which is both parameter free and provably better than a random search. It is a good alternative to Bayesian optimization methods [77], that typically require the definition of prior assumptions about the function being optimized and thus require domain knowledge.

IED name	**Instantaneous rate** $\lambda(\tau)$	**Max rate** $\lambda_{\max}$	**MOSAIC [34] implementation**	**Laplace mixture** $p(\ell)$ **[32]** such that $\psi(\tau) = \int_0^\infty \ell e^{-\ell\tau}\, p(\ell)\, \mathrm{d}\ell$
Exponential	$\lambda(\tau) = \lambda$	λ	fixed	$p(\ell) = \delta(\ell - \lambda)$
Weibull	$\lambda(\tau) = \beta\, \tau^{\alpha-1}$	∞	$\lambda(\tau_{\max})$ $[\alpha \geq 1]$	ℓ is positive α-stable [70, 71] with $\mathbb{E}[e^{-\ell\tau}] = e^{-(\mu\tau)^\alpha}$, $\mu = (\beta/\alpha)^{1/\alpha}$ $[\alpha < 1]$
Gamma	$\lambda(\tau) = \frac{\beta^\alpha\, \tau^{\alpha-1} e^{-\beta\tau}}{\Gamma(\alpha, \beta\tau)}$	β $[\alpha \geq 1]$	$\lambda(\tau_{\max})$ $[\alpha \geq 1]$	$p(\ell) = \frac{\beta^\alpha\, H(\ell - \beta)}{\Gamma(\alpha)\Gamma(1-\alpha)\, \ell\, (\ell - \beta)^\alpha}$ $[\alpha < 1]$
Gaussian / Normal	$\lambda(\tau) = \frac{\psi(\tau)}{S(\tau)}$	∞	$\lambda(\tau_{\max})$	✗
Lognormal	$\lambda(\tau) = \frac{\psi(\tau)}{S(\tau)}$	finite	fixed	✗
Cauchy	$\lambda(\tau) = \frac{\psi(\tau)}{S(\tau)}$	finite	fixed	✗
Power-law (Pareto)	$\lambda(\tau) = \begin{cases} \frac{\alpha}{\tau}, & \tau \geq \tau_0 \\ 0, & \tau < \tau_0 \end{cases}$	$\frac{\alpha}{\tau_0}$	fixed	✗
Power-law (Lomax)	$\lambda(\tau) = \frac{\alpha\kappa}{1 + \kappa\tau}$	$\alpha\kappa$	fixed	$p(\ell) = \frac{\ell^{\alpha-1} e^{-\ell/\kappa}}{\Gamma(\alpha)\kappa^\alpha}$

Table 3: Hazard-based and Laplace-mixture ingredients used by `NoMaSS`. For each supported inter-event time distribution (IED) with density $\psi(\tau)$, we report the instantaneous rate (hazard) $\lambda(\tau) = \psi(\tau)/S(\tau)$, where the survival function is $S(\tau) = \int_\tau^\infty \psi(u)\, du$. When a simple closed form is available, we list it explicitly; otherwise we retain the generic ratio form $\psi(\tau)/S(\tau)$ (see Table 2 for complete parametrization). We also report the maximal hazard $\lambda_{\max}$ used to set the rejection/thinning bound in MOSAIC. Here $\tau_{\max}$ denotes the current age of the *oldest* reactant (i.e., the one that has not reacted for the longest time), at which the bound is evaluated. When available, we provide the Laplace-mixture density $p(\ell)$ such that $\psi(\tau) = \int_0^\infty \ell e^{-\ell\tau}\, p(\ell)\, d\ell$, enabling simulation via the Laplace–Gillespie construction. We use ℓ for the latent mixture rate to avoid confusion with the nominal rate parameter λ.

Density estimation of mRNA distribution. Generally, considering a kernel density estimator for a random variable $Y = g(X)$ denoted $f_Y(y)$, one can retrieve the density function of X, denoted $f_X(x)$, with the relation (up to a scaling factor) [78]

$$f_X(x) \sim f_Y(y) \left| \frac{d}{dx} g(x) \right|. \qquad [7]$$

Since in our problem, the mRNA density count is bounded between $[0, +\infty]$, we cannot estimate the density directly with a gaussian kernel as it may infer non zero density for negative mRNA counts. Thus, we first transform mRNA counts with the function $y = g(x) = \log(x)$ so that the transformed data is unbounded ($[-\infty, +\infty]$). Then, we retrieve the density of the mRNA counts with

$$f_X(x) \sim \frac{f_Y(\log(x))}{x}.$$

Variance-based sensitivity analysis. A variance-based sensitivity analysis decomposes the variance of the output of the model $\mathrm{Var}(Y)$ into fractions which can be attributed to inputs $\{S_i\}$, where a first order sensitivity index is defined as

$$S_i = \frac{V_i}{\mathrm{Var}(Y)},$$

We note that with this definition, the sum of the sensitivities over all input parameters is always one ($\sum S_i = 1$). We estimate sensitivity indices numerically with a quasi-Monte Carlo method, where points are sampled with the Sobol quasi-random sequences [55]. The python library `Salib` [79] is used to perform this analysis.

Acknowledgements.

The authors thank Jonathan Karr, Farshid Jafarpour, Srividya Iyer-Biswas and Peter Ashcroft for their valuable suggestions. This research was supported by the COSMIC European Training Network, funded from the European Union's Horizon 2020 research and innovation program under grant agreement No 765158, and from the Swiss National Science Foundation (Sinergia CRSII5 193832 project).

Competing interrests.

The authors declare no competing interest

Data availability.

NoMaSS is implemented as a `python` library (`-pip install NoMaSS`), and its source code is accessible at https://github.com/AI-SysBio/NoMaSS. All datasets used in this studies are made available in the repository.

Bibliography

[1] Sheldon M Ross et al. *Stochastic processes*. Vol. 2. Wiley New York, 1996.

[2] Nicolaas Godfried Van Kampen. *Stochastic processes in physics and chemistry*. Vol. 1. Elsevier, 1992.

[3] Yang Cao and David C Samuels. "Discrete stochastic simulation methods for chemically reacting systems". In: *Methods in enzymology* 454 (2009), pp. 115–140.

[4] Ping Yan. "Distribution theory, stochastic processes and infectious disease modelling". In: *Mathematical epidemiology*. Springer, 2008, pp. 229–293.

[5] Masaaki Kijima. *Stochastic processes with applications to finance*. Chapman and Hall/CRC, 2002.

[6] Yi-Ting Tseng et al. "Forecasting the seasonal pollen index by using a hidden Markov model combining meteorological and biological factors". In: *Science of the Total Environment* 698 (2020), p. 134246.

[7] Daniel T Gillespie. "A general method for numerically simulating the stochastic time evolution of coupled chemical reactions". In: *Journal of computational physics* 22.4 (1976), pp. 403–434.

[8] Daniel T Gillespie. "Exact stochastic simulation of coupled chemical reactions". In: *The journal of physical chemistry* 81.25 (1977), pp. 2340–2361.

[9] Grazziela P Figueredo et al. "Comparing stochastic differential equations and agent-based modelling and simulation for early-stage cancer". In: *PloS one* 9.4 (2014), e95150.

[10] Adam Arkin, John Ross, and Harley H McAdams. "Stochastic kinetic analysis of developmental pathway bifurcation in phage λ-infected Escherichia coli cells". In: *Genetics* 149.4 (1998), pp. 1633–1648.

[11] Maria Rodriguez Martinez et al. "Messenger RNA fluctuations and regulatory RNAs shape the dynamics of a negative feedback loop". In: *Physical Review E* 81.3 (2010), p. 031924.

[12] Marcel Jan Thomas et al. "A probabilistic model of the germinal center reaction". In: *Frontiers in immunology* 10 (2019), p. 689.

[13] Aurélien Pélissier et al. "Computational model reveals a stochastic mechanism behind germinal center clonal bursts". In: *Cells* 9.6 (2020), p. 1448.

[14] Christian L Vestergaard and Mathieu Génois. "Temporal gillespie algorithm: Fast simulation of contagion processes on time-varying networks". In: *PLoS computational biology* 11.10 (2015).

[15] Lei Sun et al. "Stochastic gene expression influences the selection of antibiotic resistance mutations". In: *Molecular biology and evolution* 37.1 (2020), pp. 58–70.

[16] Gregory AL White et al. "Demonstration of non-Markovian process characterisation and control on a quantum processor". In: *Nature Communications* 11.1 (2020), pp. 1–10.

[17] Thomas Guérin, Olivier Bénichou, and Raphaël Voituriez. "Non-Markovian polymer reaction kinetics". In: *Nature chemistry* 4.7 (2012), pp. 568–573.

[18] Cihan Ayaz et al. "Non-Markovian modeling of protein folding". In: *Proceedings of the National Academy of Sciences* 118.31 (2021), e2023856118.

[19] Hadrien Vroylandt et al. "Likelihood-based non-Markovian models from molecular dynamics". In: *Proceedings of the National Academy of Sciences* 119.13 (2022), e2117586119.

[20] Benjamin Lickert and Gerhard Stock. "Modeling non-Markovian data using Markov state and Langevin models". In: *The Journal of Chemical Physics* 153.24 (2020), p. 244112.

[21] Dmitri Bratsun et al. "Delay-induced stochastic oscillations in gene regulation". In: *Proceedings of the National Academy of Sciences* 102.41 (2005), pp. 14593–14598.

[22] Patrick S Stumpf et al. "Stem cell differentiation as a non-Markov stochastic process". In: *Cell Systems* 5.3 (2017), pp. 268–282.

[23] Zhixing Cao and Ramon Grima. "Analytical distributions for detailed models of stochastic gene expression in eukaryotic cells". In: *Proceedings of the National Academy of Sciences* 117.9 (2020), pp. 4682–4692.

[24] Roland Baddeley et al. "Responses of neurons in primary and inferior temporal visual cortices to natural scenes". In: *Proceedings of the Royal Society of London. Series B: Biological Sciences* 264.1389 (1997), pp. 1775–1783.

[25] Zhi-Qiang Jiang et al. "Calling patterns in human communication dynamics". In: *Proceedings of the National Academy of Sciences* 110.5 (2013), pp. 1600–1605.

[26] Kun Zhao and Ginestra Bianconi. "Social interactions model and adaptability of human behavior". In: *Frontiers in physiology* 2 (2011), p. 101.

[27] Albert-Laszlo Barabasi. "The origin of bursts and heavy tails in human dynamics". In: *Nature* 435.7039 (2005), pp. 207–211.

[28] Alvaro Corral. "Long-term clustering, scaling, and universality in the temporal occurrence of earthquakes". In: *Physical Review Letters* 92.10 (2004), p. 108501.

[29] Jiajun Zhang and Tianshou Zhou. "Markovian approaches to modeling intracellular reaction processes with molecular memory". In: *Proceedings of the National Academy of Sciences* 116.47 (2019), pp. 23542–23550.

[30] Lawrence Rabiner and Biinghwang Juang. "An introduction to hidden Markov models". In: *ieee assp magazine* 3.1 (1986), pp. 4–16.

[31] Byung-Jun Yoon. "Hidden Markov models and their applications in biological sequence analysis". In: *Current genomics* 10.6 (2009), pp. 402–415.

[32] Naoki Masuda and Luis EC Rocha. "A Gillespie algorithm for non-Markovian stochastic processes". In: *SIAM Review* 60.1 (2018), pp. 95–115.

[33] Marian Boguná et al. "Simulating non-Markovian stochastic processes". In: *Physical Review E* 90.4 (2014), p. 042108.

[34] Aurélien Pélissier et al. "Unifying non-Markovian dynamics and agent heterogeneity in scalable stochastic networks". In: *Nature Communications* (2026).

[35] Manuel Barrio et al. "Oscillatory regulation of Hes1: discrete stochastic delay modelling and simulation". In: *PLoS computational biology* 2.9 (2006), e117.

[36] Xiaodong Cai. "Exact stochastic simulation of coupled chemical reactions with delays". In: *The Journal of chemical physics* 126.12 (2007).

[37] David F Anderson. "A modified next reaction method for simulating chemical systems with time dependent propensities and delays". In: *The Journal of chemical physics* 127.21 (2007).

[38] Rajesh Ramaswamy and Ivo F Sbalzarini. "A partial-propensity formulation of the stochastic simulation algorithm for chemical reaction networks with delays". In: *The Journal of chemical physics* 134.1 (2011).

[39] Xiaoming Fu et al. "DelaySSAToolkit. jl: stochastic simulation of reaction systems with time delays in Julia". In: *Bioinformatics* 38.17 (2022), pp. 4243–4245.

[40] Srividya Iyer-Biswas et al. "Scaling laws governing stochastic growth and division of single bacterial cells". In: *Proceedings of the National Academy of Sciences* 111.45 (2014), pp. 15912–15917.

[41] Jue D Wang and Petra A Levin. "Metabolism, cell growth and the bacterial cell cycle". In: *Nature Reviews Microbiology* 7.11 (2009), pp. 822–827.

[42] Jennifer C England et al. "Global regulation of gene expression and cell differentiation in Caulobacter crescentus in response to nutrient availability". In: *Journal of bacteriology* 192.3 (2010), pp. 819–833.

[43] Alejandro Sánchez Alvarado and Shinya Yamanaka. "Rethinking differentiation: stem cells, regeneration, and plasticity". In: *Cell* 157.1 (2014), pp. 110–119.

[44] Ian Chambers et al. "Nanog safeguards pluripotency and mediates germline development". In: *Nature* 450.7173 (2007), pp. 1230–1234.

[45] Michael B Elowitz et al. "Stochastic gene expression in a single cell". In: *Science* 297.5584 (2002), pp. 1183–1186.

[46] Vahid Shahrezaei and Peter S Swain. "Analytical distributions for stochastic gene expression". In: *Proceedings of the National Academy of Sciences* 105.45 (2008), pp. 17256–17261.

[47] Amit Zeisel et al. "Coupled pre-mRNA and mRNA dynamics unveil operational strategies underlying transcriptional responses to stimuli". In: *Molecular systems biology* 7.1 (2011), p. 529.

[48] David M Suter et al. "Mammalian genes are transcribed with widely different bursting kinetics". In: *science* 332.6028 (2011), pp. 472–474.

[49] Srividya Iyer-Biswas, Fernand Hayot, and Ciriyam Jayaprakash. "Stochasticity of gene products from transcriptional pulsing". In: *Physical Review E* 79.3 (2009), p. 031911.

[50] Anton JM Larsson et al. "Genomic encoding of transcriptional burst kinetics". In: *Nature* 565.7738 (2019), pp. 251–254.

[51] Jong Kyoung Kim and John C Marioni. "Inferring the kinetics of stochastic gene expression from single-cell RNA-sequencing data". In: *Genome biology* 14.1 (2013), pp. 1–12.

[52] Seong Jun Park et al. "The chemical fluctuation theorem governing gene expression". In: *Nature communications* 9.1 (2018), p. 297.

[53] Yulei Wang et al. "Precision and functional specificity in mRNA decay". In: *Proceedings of the National Academy of Sciences* 99.9 (2002), pp. 5860–5865.

[54] Nacho Molina et al. "Stimulus-induced modulation of transcriptional bursting in a single mammalian gene". In: *Proceedings of the National Academy of Sciences* 110.51 (2013), pp. 20563–20568.

[55] Ilya M Sobol. "Global sensitivity indices for nonlinear mathematical models and their Monte Carlo estimates". In: *Mathematics and computers in simulation* 55.1-3 (2001), pp. 271–280.

[56] Zihao Wang, Zhenquan Zhang, and Tianshou Zhou. "Analytical results for non-markovian models of bursty gene expression". In: *Physical Review E* 101.5 (2020), p. 052406.

[57] Alexander Saichev and Didier Sornette. "Theory of earthquake recurrence times". In: *Journal of Geophysical Research: Solid Earth* 112.B4 (2007).

[58] Alexei Vazquez et al. "Impact of non-Poissonian activity patterns on spreading processes". In: *Physical review letters* 98.15 (2007), p. 158702.

[59] Anzhi Sheng et al. "Constructing temporal networks with bursty activity patterns". In: *Nature Communications* 14.1 (2023), p. 7311.

[60] Taro Takaguchi and Naoki Masuda. "Voter model with non-Poissonian interevent intervals". In: *Physical Review E—Statistical, Nonlinear, and Soft Matter Physics* 84.3 (2011), p. 036115.

[61] Kevin R Sanft and Hans G Othmer. "Constant-complexity stochastic simulation algorithm with optimal binning". In: *The Journal of chemical physics* 143.7 (2015), 08B609_1.

[62] Jarosław Gil and Andrzej Polański. "Application of Gillespie algorithm for simulating evolution of fitness of microbial population". In: *Applied Computer Science* 18.4 (2022), pp. 5–15.

[63] Ulysse Herbach et al. "Inferring gene regulatory networks from single-cell data: a mechanistic approach". In: *BMC systems biology* 11.1 (2017), pp. 1–15.

[64] Onno Boxma et al. "On/off storage systems with state-dependent input, output, and switching rates". In: *Probability in the Engineering and Informational Sciences* 19.1 (2005), pp. 1–14.

[65] Khashayar Pakdaman, Michele Thieullen, and Gilles Wainrib. "Fluid limit theorems for stochastic hybrid systems with application to neuron models". In: *Advances in Applied Probability* 42.3 (2010), pp. 761–794.

[66] Stefan Hoops et al. "COPASI—a complex pathway simulator". In: *Bioinformatics* 22.24 (2006), pp. 3067–3074.

[67] Petter Holme and Jari Saramäki. "Temporal networks". In: *Physics reports* 519.3 (2012), pp. 97–125.

[68] Nicola Perra et al. "Activity driven modeling of time varying networks". In: *Scientific reports* 2.1 (2012), p. 469.

[69] Samuel Cure, Florian G Pflug, and Simone Pigolotti. "Fast and exact stochastic simulations of epidemics on static and temporal networks". In: *PLOS Computational Biology* 21.9 (2025), e1013490.

[70] Howard Pollard. "Representation of e^{-x^λ} As a Laplace Integral". In: *Bulletin of the American Mathematical Society* 52 (1946), p. 908.

[71] Marek Kanter. "Stable densities under change of scale and total variation inequalities". In: *The Annals of Probability* (1975), pp. 697–707.

[72] Yuxin Deng and Wenjie Du. "The Kantorovich metric in computer science: A brief survey". In: *Electronic Notes in Theoretical Computer Science* 253.3 (2009), pp. 73–82.

[73] Yossi Rubner, Carlo Tomasi, and Leonidas J Guibas. "A metric for distributions with applications to image databases". In: *Sixth International Conference on Computer Vision (IEEE Cat. No. 98CH36271)*. IEEE. 1998, pp. 59–66.

[74] Nicolas Bonneel et al. "Displacement interpolation using Lagrangian mass transport". In: *ACM Transactions on Graphics (TOG)*. Vol. 30. 6. ACM. 2011, p. 158.

[75] Ofir Pele and Michael Werman. "Fast and robust earth mover's distances". In: *2009 IEEE 12th international conference on computer vision*. IEEE. 2009, pp. 460–467.

[76] King Davis. "A Global Optimization Algorithm Worth Using". In: *dlib C++ library*. 2017. URL: http://blog.dlib.net/2017/12/a-global-optimization-algorithm-worth.html.

[77] Jasper Snoek, Hugo Larochelle, and Ryan P Adams. "Practical bayesian optimization of machine learning algorithms". In: *Advances in neural information processing systems*. 2012, pp. 2951–2959.

[78] Matthew P Wand, James Stephen Marron, and David Ruppert. "Transformations in density estimation". In: *Journal of the American Statistical Association* 86.414 (1991), pp. 343–353.

[79] Jon Herman and Will Usher. "SALib: an open-source Python library for sensitivity analysis". In: *Journal of Open Source Software* 2.9 (2017), p. 97.

Supplementary Materials

Gillespie-Based Simulation and Inference for Non-Markovian Stochastic Reaction Networks

Aurelien Pelissier[1,2,3], **Miroslav Phan**[1,2], **Niko Beerenwinkel**[2], **Marıa Rodrıguez Martınez**[1,5]

[1]IBM Research Europe, 8803 Ruschlikon, Switzerland
[2]Department of Biosystems Science and Engineering, ETH Zurich, 4058 Basel, Switzerland
[3]Institute of Computational Life Sciences, Zurich University of Applied Sciences (ZHAW), 8820, Wadenswil, Switzerland
[4]Department of Biomedical Informatics & Data Science, Yale School of Medicine, New Haven, CT, United States
[†]Corresponding author maria.rodriguezmartinez@yale.edu.

A Non-Markovian division

A.1 Cell inter-division time

On Figure S1 and Table S1, we show the normal, Weibull, gamma and log-normal distributions with parameters chosen to fit experimentally measured cell cycle inter event times for *Bacillus subtilis* [1] and *Escherichia coli* [2] (https://jun.ucsd.edu/repository.php). The log-normal distribution provides the optimal fit in all three cases as it minimizes the Earth Mover's Distance (EMD) [3].

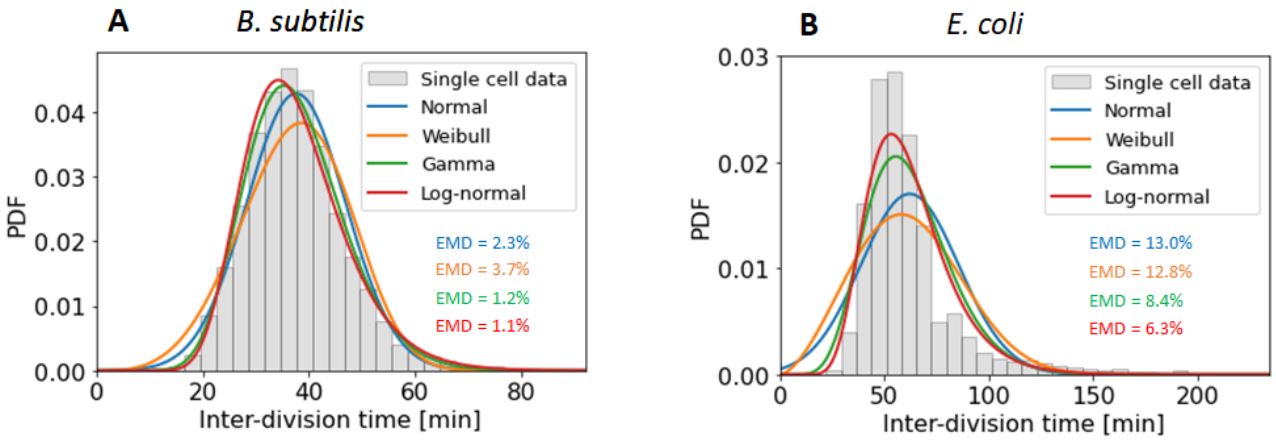


Figure S1: Measured interdivision time of (A) *Bacillus subtilis* [1] and (B) *Escherichia coli* [2] at constant temperature fitted with a normal, log-normal, gamma and Weibull distribution. The fitted distributions are quantified in term of the Earth Mover's Distance (EMD) to the measured single cell inter-division distribution.

Distribution	Exponential	Normal	Log-normal	Gamma	Weibull
EMD - *Bacillus subtilis [1]*	55.8	2.3	**1.1**	**1.2**	3.7
Optimal parameter ($1/\lambda_0$)	37.6 min	37.6 min	37.6 min	37.6 min	37.4 min
Optimal parameter (γ or α)	–	0.25	0.26	16.4	4.2
EMD - *Escherichia coli [2]*	47.4	13.0	**6.3**	8.4	12.8
Optimal parameter ($1/\lambda_0$)	62.0 min	62.0 min	61.7 min	62.0 min	61.8 min
Optimal parameter (γ or α)	–	0.38	0.32	9.3	2.6
EMD - *Caulobacter crescentus [4]*	64.9	2.1	**1.8**	**1.7**	5.0
Optimal parameter ($1/\lambda_0$)	72.1 min	72.1 min	72.1 min	72.1 min	71.4 min
Optimal parameter (γ or α)	–	0.12	0.12	70.8	7.1

Table S1: Fitting performance of various distributions to measured single cell inter-division time distribution of three types of bacteria dividing at constant temperature. The optimal parameters as well as the computed Earth Mover Distance (EMD) between the fitted and measured distributions are provided. EMDs are given with a confidence interval of $\pm 10\,\%$ of the provided value, due to stochaticity in the fitting process. Lowest EMD scores for each type of bacteria are highlighted in bold.

A.2 Symmetric division

A simple system following non-Markovian dynamics that can be described analytically is symmetric division, where a single type of reactant A reacts to produce $\nu > 1$ identical versions of itself. We thus consider the reaction $A \to \nu A$. In the general case, the population's steady state exponential growth rate k is related to the single-cell division time distribution $\psi(t)$ through the integral [5]

$$\mathcal{L}\{\psi\}(k) = \int_0^\infty e^{-k\tau}\psi(\tau)d\tau = \frac{1}{\nu}. \qquad [8]$$

In words, k is the point at which the Laplace transform of the division time distribution $\psi(\tau)$ is equal to $1/\nu$. Then, the mean doubling time of the population is obtained with $t_2 = \ln(2)/k$. Defining the mean division time of single cells as $\mu = \int_0^\infty t\psi(t)dt$, we can solve the above equation for particular cases.

Deterministic processes. For a deterministic division (Dirac distribution), with PDF $\psi(t) = \delta(t-\mu)$, the Laplace transform is given by

$$\mathcal{L}\{\psi\}(k) = \exp(-\mu k). \qquad [9]$$

Thus, we can simplify equation 8 into

$$\exp\left(-\frac{\ln(2)\cdot\mu}{t_2}\right) = \frac{1}{\nu} \text{ which yields } \boxed{t_2 = \mu\cdot\frac{\ln(2)}{\ln(\nu)}}. \qquad [10]$$

Markovian processes. For an exponential distribution, with PDF $\psi(t) = \lambda e^{-\lambda t}$, the Laplace transform is given by

$$\mathcal{L}\{\psi\}(k) = \frac{\lambda}{k+\lambda}. \qquad [11]$$

Replacing $\lambda = 1/\mu$, we can simplify equation 8 into

$$\frac{1}{1+\mu\ln(2)/t_2} = \frac{1}{\nu} \text{ which yields } \boxed{t_2 = \mu\cdot\frac{\ln(2)}{\nu-1}}. \qquad [12]$$

Non-Markovian processes. From Eq.8, the population doubling time can be computed exactly for any distributions with an analytical form of its Laplace transform. We provide an example with the gamma distribution, with PDF $\psi(t) = \frac{\beta^\alpha}{\Gamma(\alpha)}t^{\alpha-1}e^{-\beta t}$ and Laplace transform

$$\mathcal{L}\{\psi\}(k) = \frac{\beta^\alpha}{(\beta+k)^\alpha}. \qquad [13]$$

Replacing $\beta = {}^{\alpha}/_{\mu}$, we can simplify equation 8 into

$$\frac{1}{1+\mu\ln(2)/(\alpha t_2)} = \left(\frac{1}{\nu}\right)^{1/\alpha} \quad \text{which yields} \quad \boxed{t_2 = \mu \cdot \frac{\ln(2)}{\alpha\left(\nu^{1/\alpha}-1\right)}} \qquad [14]$$

where $\alpha = 1$ recovers the Markovian case, and $\alpha = \infty$ the deterministic case[1]. We illustrate these result Figure S2, where the influence of the shape parameter on the doubling rate can be visualized. We simulate the non-Markovian division process with NoMaSS and fit the obtained population growth with an exponential (which result in a straight line when plotting with a logarithmic scale, Figure S2B). As expected from the analytical expression Eq.14, the obtained growth rate reduces with the shape parameter and, in the special case of $\nu = 2$, converge to $k = \ln 2$ for increasing α.

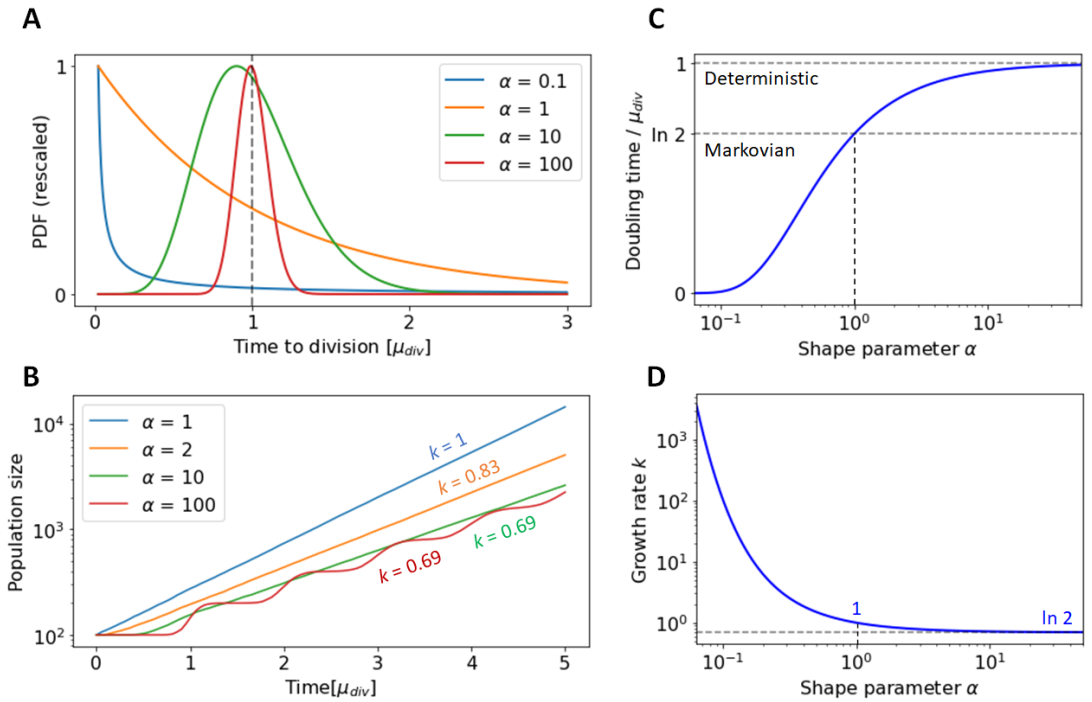


Figure S2: Population dynamics of Non-Markovian division following the equation $A \to 2A$ with a single cell interdivision time distribution following a gamma distribution of shape parameter α and mean μ_{div}. (A) PDF of gamma distributions with mean 1 and shape parameter α, rescaled for visual clarity. (B) Population growth of A for different shape parameter simulated with NoMaSS. The fitted exponential growth rate (k) is provided above each line. (C) Analytical ratio between the population doubling time ($t_2 = \ln(2)/k$) and the single cell average division time for different value of α. (D) Analytical growth rate of the population size for different value of α. Special cases for the Markovian ($\alpha = 1$) and deterministic ($\alpha \to \infty$) divisions are annotated.

A.3 Asymmetric division

A more complex non-Markovian system, that we describe in the main text, is asymmetric division, where a reactant A reacts to produce ν_A versions of itself as well as ν_B versions of another reactant B, following the reaction (i) $A \to \nu_A A + \nu_B B$. Then, reactant B does not divide but can differentiate into A following the reaction (ii) $B \to A$. In this setup, it was shown analytically that the population of both reactants A and B grow exponentially with the same rate k for any inter-event distribution of reactions (i) and (ii), where the growth rate k is related to the parameters ν_A and ν_B with the relation [5]

$$\mathcal{L}\{\psi_{\text{div}}\}(k) = \frac{1}{\nu_A + \nu_B \cdot \mathcal{L}\{\psi_{\text{diff}}\}(k)}, \qquad [15]$$

[1]Applying l'Hôpital's rule, It can be shown that $\lim_{x\to\infty}(xa^{1/x} - x) = \ln(a)$

where ψ_{div} and ψ_{diff} denote the PDF of reaction (i) and (ii), respectively. In the Markovian case, one can solve this equation with:

$$\frac{\lambda_{\text{div}}}{k+\lambda_{\text{div}}} = \frac{k+\lambda_{\text{diff}}}{\nu_A(k+\lambda_{\text{diff}})+\nu_B\lambda_{\text{diff}}}, \quad [16]$$

which yields the quadratic equation

$$k^2 + k\cdot(\lambda_{\text{div}} - \nu_A\lambda_{\text{div}} + \lambda_{\text{diff}}) + \lambda_{\text{div}}\lambda_{\text{diff}}(1-\nu_A-\nu_B) = 0, \quad [17]$$

where we take the positive solution for the growth rate

$$k = \frac{1}{2}\left(\lambda_{\text{div}}\nu_A - \lambda_{\text{div}} - \lambda_{\text{diff}} + \sqrt{(\lambda_{\text{diff}}+\lambda_{\text{div}}-\lambda_{\text{div}}\nu_A)^2 + 4\lambda_{\text{div}}\lambda_{\text{diff}}(\nu_A+\nu_B-1)}\right). \quad [18]$$

Unconstrained microbial growth of *Caulobacter crescentus*. We now consider the specific setting of asymmetric division with $\nu_A = \nu_B = 1$ (Figure S4A), corresponding to the unconstrained microbial growth of *Caulobacter crescentus*. In the Markovian case, we can write

$$k = \frac{\lambda_{\text{diff}}}{2}\left(\sqrt{1+4\frac{\lambda_{\text{div}}}{\lambda_{\text{diff}}}} - 1\right). \quad [19]$$

In the limit $\lambda_{\text{div}} \ll \lambda_{\text{diff}}$, using the Taylor expansion $\sqrt{1+x} \sim 1 + x/2$, we obtain $k = \lambda_{\text{div}}$, which recovers the symmetric case ($A \to 2A$). In this regime, differentiation ($A \to B$) is very fast compared to division, so that it can be considered to occur instantly. On the other hand, in the limit where differentiation is very slow compared to division ($\lambda_{\text{div}} \gg \lambda_{\text{diff}}$), the growth rate is given by $k = \sqrt{\lambda_{\text{div}}\lambda_{\text{diff}}}$. Finally, in the specific case of $\lambda_{\text{div}} = \lambda_{\text{diff}} = \lambda$, we obtain $k = \lambda/\varphi$ where $\varphi = \frac{1+\sqrt{5}}{2}$ is the golden ratio.

In the general non-Markovian case (i.e. gamma or Dirac IEDs), there is no simple analytical solutions for k. Thus, we use NoMaSS to show how the growth rate varies (Figure S3) for different combinations of distribution parameters $(\lambda_{\text{div}}, \alpha_{\text{div}})$ and $(\lambda_{\text{diff}}, \alpha_{\text{diff}})$. As expected, changes in the mean of the distribution have a much larger

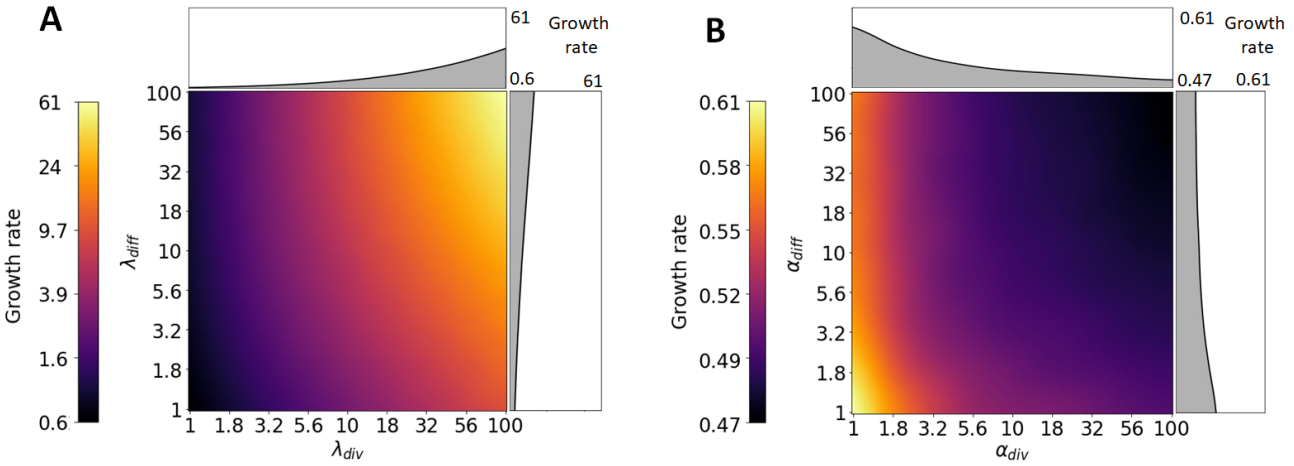


Figure S3: Growth rate of the number of cells $(N_A + N_B)$ in a system of asymmetric division ($A \to A + B$ and $B \to A$), where the two processes are modeled as a gamma distribution with shape parameter $(\lambda_{\text{div}}, \alpha_{\text{div}})$ and $(\lambda_{\text{diff}}, \alpha_{\text{diff}})$. (A) Heatmap of the growth rate as a function of the processes rate $(\lambda_{\text{div}}, \lambda_{\text{diff}})$ where $\alpha_{\text{div}} = \alpha_{\text{diff}} = 1$ are kept constant. Note that growth rate axes are in logarithmic scales (B) Heatmap of the growth rate as a function of the processes shape parameters $(\alpha_{\text{div}}, \alpha_{\text{diff}})$ where $\lambda_{\text{diff}} = \lambda_{\text{div}} = 1$ are kept constant. The projection of the growth rate averaged over the differentiation and division parameter axis are plotted on the right and the top of the heatmap, respectively.

impact on the growth rate than changes in the shape parameter. Nevertheless, the shape parameter can affect the growth rate by more than ∼20%, highlighting the importance of properly accounting for it. Interestingly, we observe that α_{div} and α_{diff} affects the growth rate in a similar manner.

Oscillation amplitudes of quiescent cells. After fitting the model parameters with experimental data from [4], consisting of single-cell measurements of division times, as well as steady-state cell populations ratios of stalked and quiescent cells at 31 °C, we were able to constrain 3 out of 4 model parameters (Figure S4B,C).

However, we were not able to constrain α_{diff} as the data only provides information about steady-state levels for quiescent cell differentiation – for technical reasons, the direct experimental characterization of quiescent cell differentiation is challenging [6]. As previously shown [5], the deterministic modeling of quiescent cell differentiation reveals transient oscillations, as the initial populations are highly synchronized and differentiate at the same time. Stochastic models should partially smooth these transient oscillations, as division and differentiation occur probabilistically, which results in the gradual loss of synchronization. However, we see that in the Markovian case the oscillations have completely disappeared (Figure S4D), i.e. the exponentially distributed inter-event times fully mix division and differentiation times destroying the synchronization. Opposed to this, the non-Markovian simulation still shows transient oscillations (Figure S4E). Interestingly, the amplitude of the quiescent cell population oscillations increases as the variance of the inter-event differentiation time of stalked cells reduces (Figure S4F). We note that with our parametrization, α_{diff} inversely correlate to the variance of the distribution through the relation $Var = 1/(\alpha_{\text{diff}}\lambda_{\text{diff}}^2)$.

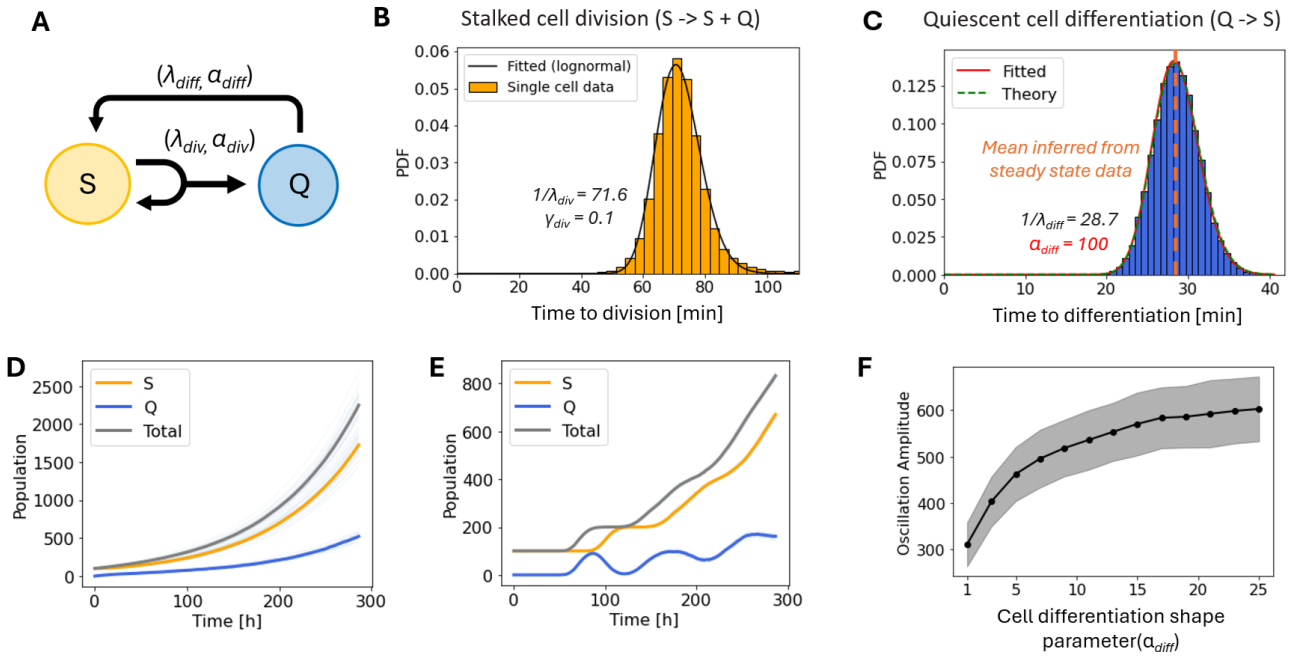


Figure S4: Microbial growth of *Caulobacter crescentus*. (A) Cartoon depicting the dynamics of microbial growth. A stalked cell S divides asymmetrically into another stalked cell and a motile quiescent cell Q that cannot divide further but can differentiate into a stalked cell. (B) Inter-event time distribution of stalked cell divisions using single-cell measurements from [4]. Data has been fitted to a log-normal distribution. The fitted distribution was later used in the NoMaSS simulation. (C) Simulated inter-event time of quiescent cell differentiation with a gamma distribution of shape parameter $\alpha_{\text{diff}} = 100$, chosen arbitrarily for the purpose of illustration (as there is no experimental data available). (D,E) Simulated population dynamics with an initial population of 100 stalked cells using Markovian processes and non-Markovian dynamics, respectively, both with the same mean inter-event time, and with the inter-event distributions taken from subplots B & C for the non-Markovian case. (F) Evolution of the oscillation amplitudes as a function of α_{diff} while keeping all other parameters constant (as shown on subplots B & C). The black line shows the mean of 1000 simulations, and the shaded area corresponds to one standard deviation. The oscillations are quantified in terms of the root mean square deviation (RMSD) between the quiescent cells population dynamics and its exponential fit.

B Modeling stem cell differentiation

This section refers to the modeling of mouse stem cell differentiation [7] dynamics. In this experiment, cells are profiled at the single-cell level at 3 different stages of differentiation (denoted A,B,C), thus undergoing two transitions ($A \to B$ and $B \to C$). For consistency across timepoints, measured population of N_A, N_B and N_C are transformed in term of proportion $(p_A, p_B, p_C) \in [0,1]^3$ following

$$p_A = \frac{N_A}{N_A + N_B + N_C}, \quad p_B = \frac{N_B}{N_A + N_B + N_C} \quad \text{and} \quad p_C = \frac{N_C}{N_A + N_B + N_C}. \tag{20}$$

As the population dynamics is described by proportions bounded between 0 and 1. This may bias the the optimization of the root mean square deviation (RMSD) towards ignoring variations close to very low (~ 0) and very high (~ 1) values, as a proportion of 98 % will be considered closer to 99 % than 50 % is to 52 %. To correct for this bias, we transform both simulated and measured populations with an inverse sigmoid (or logit) function (Figure S5). For experimental reasons, proportions lower than 1% or higher than 99% are considered not reliable enough (only a few hundred cells in experimental data), so we only apply the transformation within that range:

$$\begin{aligned} y_{\text{new}} &= \log\left(\frac{y}{1-y}\right) \quad \text{for} \quad 1\,\% \leq y \leq 99\,\% \\ y_{\text{new}} &= 4.6 \quad \text{for} \quad y > 99\,\% \\ y_{\text{new}} &= -4.6 \quad \text{for} \quad y < 1\,\% \end{aligned} \tag{21}$$

With NoMaSS, we model the two differentiations processes with different choices of inter-event time distribution, and we provide in Table S2 the distribution parameters of the optimal fit to both the raw (unstransformed) and logit transformed experimental data. Optimal parameters were obtained by minimizing the RMSD between the measured and simulated populations at each time point. Minimization was performed with the `maxLIPO` algorithm [8], which is both parameter free and provably better than a random search. Unsurprisingly, the exponential distribution consistently yields to the highest RMSD, while the normal, log-normal and gamma output similarly low RMSD. On the other hand, the Weibull distribution consistently output higher RMSD than other distributions for both the untransformed and transformed data.

Distribution	Exponential	Normal	Log-normal	Gamma	Weibull
RMSD (sigmoid transformed data)	2.1	0.38	0.39	**0.35**	0.69
Optimal parameter ($1/\lambda_1$)	29.4 h	31.9 h	32.3 h	31.9 h	32.2 h
Optimal parameter ($1/\lambda_2$)	66.7 h	66.7 h	62.9 h	65.4 h	63.3 h
Optimal parameter (γ_1 or α_1)	–	0.24	0.21	20.2	4.3
Optimal parameter (γ_2 or α_2)	–	0.15	0.15	38.7	4.8
RMSD (untransformed data)	0.22	0.017	**0.014**	0.019	0.039
Optimal parameter ($1/\lambda_1$)	31.9 h	32.1 h	31.6 h	29.5 h	31.8 h
Optimal parameter ($1/\lambda_2$)	63.7 h	59.2 h	59.9 h	61.0 h	58.8 h
Optimal parameter (γ_1 or α_1)	–	0.27	0.27	27.6	4.65
Optimal parameter (γ_2 or α_2)	–	0.050	0.083	39.3	4.94

Table S2: Root mean square deviation (RMSD) between simulated and measured stem cell differentiation data for both the raw and sigmoid transformed data. Lowest RMSD scores are highlighted in bold.

C Non-Markovian dynamics of RNA transcription

This section provide additional information regarding the two steps model of RNA transcription described in the main text. As a reminder, we consider a promoter switching between an ON and an OFF state with gamma inter-event distributions parametrized by $(\lambda_{\text{on}}, \alpha_{\text{on}})$ and $(\lambda_{\text{off}}, \alpha_{\text{off}})$. Some mRNA may be produced only if the promoter state is ON, with a gamma inter-event distribution $(\lambda_{\text{prod}}, \alpha_{\text{prod}})$. The mRNA then degrades at a constant rate λ_{deg}. The system is initialized with a promoter in an OFF state and a count of 0 mRNA.

C.1 Oscillations of mRNA dynamics

Modeling RNA transcription with non-Markovian processes may results into predictable oscillations of averaged mRNA counts before the steady state is reached, a phenomenon that is not observed in the Markovian case.

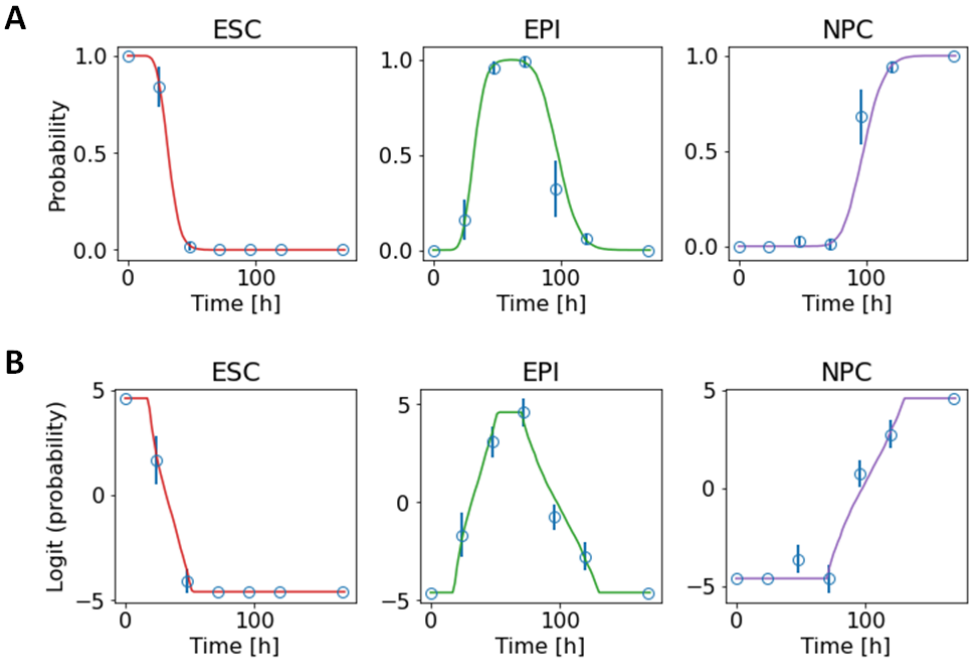


Figure S5: Normalized stem cell population as a function of time for the three stages of differentiations following the reactions ESC → EPI and EPI → NPC. The blue points represent the mean and standard deviation of the stem cell populations measured at different time points [7], and the solid lines depict the population dynamics averaged over 1000 NoMaSS simulation, with differentiation modelled as the best fitting gamma distribution. Results are shown for (A) the untransformed and (B) transformed probabilities with an inverse sigmoid (or logit) function.

For example, in the specific case of a deterministic promoter transition (promoter activation and deactivation modeled as Dirac distributions), the promoter oscillates between the G_{on} and G_{off} state indefinitely with a period of $T = 1/\lambda_{on} + 1/\lambda_{off}$, which results in oscillations in mRNA count with the same period. Adding some stochaticity to the promoter transition results in a decay of the G_{on} and G_{off} probability oscillation amplitude, until it converges to the steady state value and no more amplitudes can be observed (Figure S6A&B). Interestingly, the more *stochasticity* is added to the promoter transition reaction (decreasing shape parameter α_{on} and α_{off}), the faster the oscillation decays (Figure S6B&C). In the case of Markovian mRNA transcription ($\alpha_{on} = \alpha_{off} = 1$), the system reaches directly its steady state after a single period T and oscillations are not observed (Figure S6A).

C.2 Modeling the reaction to stimuli in RNA transcription

Non steady state RNA transcription dynamics are not straight forward to observe directly experimentally. A possibility is to study the behavior of RNA dynamics after stimuli by the epidermal growth factor (EGF) or lipopolysaccharide (LPS), which initiate well characterized signaling cascades that culminate in orchestrated transcriptional responses [9]. After the stimuli, the system take some times to reach its new steady state, which results in observable non-steady state dynamics shortly after the stimuli. It was shown that the stimuli may affect all rates involved in RNA transcription ($\lambda_{on}, \lambda_{off}, \lambda_{deg}$) [9] (Figure S7B). To avoid introducing too many new parameters in our model to incorporate the stimuli, we keep λ_{deg} and λ_{off} constant, and we set the promoter activation rate $\lambda_{on}(t)$ to change with time as a function of two terms:

$$\lambda_{on}(t) = \lambda_{on}^{decay}(t) + \lambda_{on}^{peak}(t). \qquad [22]$$

The first one is a progressive decay (modeled with a sigmoid function) from its value before the stimuli (λ_A) to its new value (λ_B) as

$$\lambda_{on}^{decay}(t) = (\lambda_A - \lambda_B) \cdot \frac{e^{-b\,(t-t_0)}}{1 + e^{-b\,(t-t_0)}} + \lambda_B \qquad [23]$$

where t refers to the time elapsed since the stimuli. Additionally, we also add a second term to simulate the short term peak induced by the stimuli, as introduced in original study [9]:

$$\lambda_{on}^{peak}(t) = a_1 \frac{(t/t_p)^{n_1}}{(1 + t/t_p)^{n_2}}. \text{ with } n_1 < n_2 \qquad [24]$$

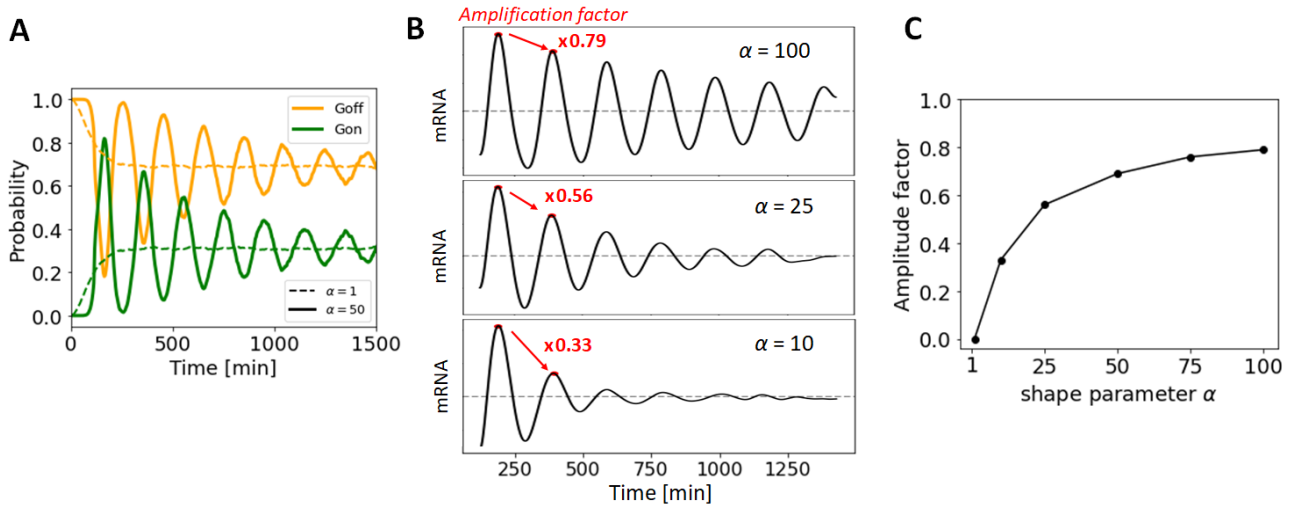


Figure S6: Oscillations in non-Markovian mRNA transcription dynamics where we set $\alpha = \alpha_{\text{on}} = \alpha_{\text{off}}$ for simplicity. (A) Time dependent probability that the promoter is in the on or off states for both the Markovian ($\alpha = 1$, dashed line) and non Markovian case ($\alpha = 50$, solid line) at each time point, obtained after averaging over 10k simulations. (B) Time evolution of mRNA count averaged over 10k NoMaSS simulations for different shape parameters α. For visual clarity, only the time interval where the mRNA count oscillate around its steady state value (dashed line) is plotted. The observed factor at which the oscillation amplitude decays between each period is shown is red. (C) Dependency of the shape parameter α on the oscillation amplification factor (amplitude of the second peak divided by the amplitude of the first peak) of averaged mRNA counts.

This peak induce a very short spike in the λ_{on} rate (Figure S7B) and cause the system to leave its steady state to enter non-Markovian dynamics. To make the parameter optimization easier, we fix the parameters $t_p = 5\,\text{min}$, $n_1 = 2$ and $n_2 = 4$ for all our simulation. On the other hand, we leave the parameter a_1 as variable since it reflects the amplitude of the peak and thus the oscillation amplitudes of RNA transcription.

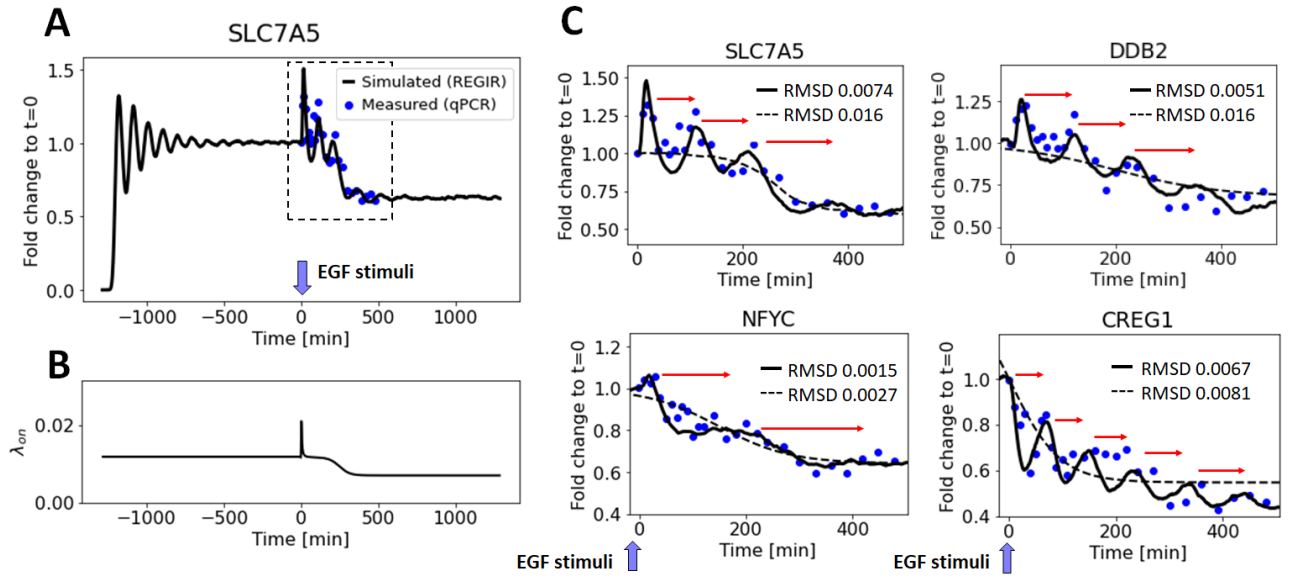


Figure S7: (A) Simulated mRNA gene expression of SLC7A5 normalized by its average expression at the time of stimulation, averaged over 10k simulation. We can clearly distinguish two phases, where the mRNA dynamics reaches first reaches its steady state and then reacts to the stimuli. (B) Time evolution of the simulated promoter activation rate of SLC7A5 (λ_{on}) before ($t < 0$) and after ($t > 0$) the EGF stimuli. (C) mRNA gene expression of different genes normalized by their average expression at the time of stimulation, where the measured expression [9] by quantitative PCR (qPCR) are compared to simulated RNA expression averaged over 10k simulations with optimal parameter fit. The simulation is provided for both NoMaSS (solid line) and the Markovian model (SG, dashed line). Red arrow highlight the increase in oscillation period caused by the decrease of λ_{on}.

The modeling of stimuli is performed by first simulating the system before the stimuli until its steady state is reached ($\lambda_{on} = \lambda_A$), and then re-initializing $t = 0$ and the promoter to its off state (Figure S7A). We adjust the models parameters to optimize the fit to experimental data for the genes CREG1, DDB2, SLC7A5, and NFYC (Figure S7C). For all simulation, we constrain $\alpha_{\mathbf{on}} = \alpha_{\mathbf{off}} = \alpha$ and $\alpha_{\mathbf{prod}} = 1$ to facilitate the parameter fit. Note that with these constraints, the non-Markovian model only have one more parameter of freedom (α) than the Markovian model. For each gene, parameters (α, λ_A, λ_B, b, t_0 and a_1) are fitted with the `maxLIPO` algorithm [8] by minimizing the RMSD between the NoMaSS output of 1000 simulations and the experimental data (Table S3). Note that since the dataset involve fold-change of mRNA since $t = 0$ rather than absolute mRNA count, it is not possible to fit parameters such as λ_{off} or λ_{prod}, so we set them to constant values for all simulations ($\lambda_{off} = 4\lambda_{deg}$ and $\lambda_{prod} = 25\lambda_{deg}$). Also, while $\lambda_{on}(t = 0) = \lambda_A$ can be inferred from the oscillation period in the non-Markovian model (NoMaSS), it is not possible to obtain it in the Markovian case (SG) as there is no oscillations (marked as X in Table S3).

Interestingly, several of the studied genes revealed oscillations in their mRNA expression after the stimuli that could not be replicated accurately by the Markovian model (e.g. RMSD = 0.016 for SLC7A5). On the other hand, NoMaSS was able to reproduce the oscillations accurately (e.g. RMSD = 0.0083 for SLC7A5). Strikingly, the genes where the most significant difference was observed between the non-Markovian and Markovian model were the one with the highest oscillation amplitudes. Another interesting aspect is the gradual increasing period of the oscillations, which indicate a progressive reduction in the promoter activation rate (red arrows on Figure S7C).

	RMSD	λ_A	λ_B	b [1/min]	t_0 [min]	a_1	α
CREG1 (SG)	0.0081	X	$0.45 \cdot \lambda_A$	0.004	0	X	|
CREG1 (NoMaSS)	**0.0067**	$\lambda_{off}/4.5$	$0.45 \cdot \lambda_A$	0.006	0	-1.2	40
DDB2 (SG)	0.016	X	$0.66 \cdot \lambda_A$	0.029	214	X	|
DDB2 (NoMaSS)	**0.0051**	$\lambda_{off}/9$	$0.66 \cdot \lambda_A$	0.012	250	1	60
NFYC (SG)	0.0027	X	$0.64 \cdot \lambda_A$	0.015	150	X	|
NFYC (NoMaSS)	**0.0015**	$\lambda_{off}/18$	$0.64 \cdot \lambda_A$	0.008	160	0.55	15
SLC7A (SG)	0.016	X	$0.63 \cdot \lambda_A$	0.041	270	X	|
SLC7A (NoMaSS)	**0.0074**	$\lambda_{off}/9.6$	$0.63 \cdot \lambda_A$	0.035	270	1.3	26

Table S3: Optimal parameters and root mean square deviation (RMSD) between the simulated and measured mRNA time resolved gene expression fold changes after EGF stimulation [9]. Results are provided for both the SG and NoMaSS model, each averaged over 1000 simulations. Best RMSD scores for each genes are highlighted in bold. Parameters marked with | indicate that it is not applicable in the model. Parameters marked with X mean that the model is insensitive to its value, so it's irrelevant in the context of optimization.

C.3 Phase diagrams and sensitivity analysis of the steady state mRNA count distribution

In the main text, we performed a variance-based sensitivity analysis of the coefficient of variation (CV = std/mean) of the steady state mRNA distribution. In Figure S8, we provide the same sensitivity analysis for the mean, and the differential entropy H. Intuitively, H measures the randomness of a random variable and the number of bits required to describe it. Formally, H is computed as a generalization of the Shannon entropy to continuous

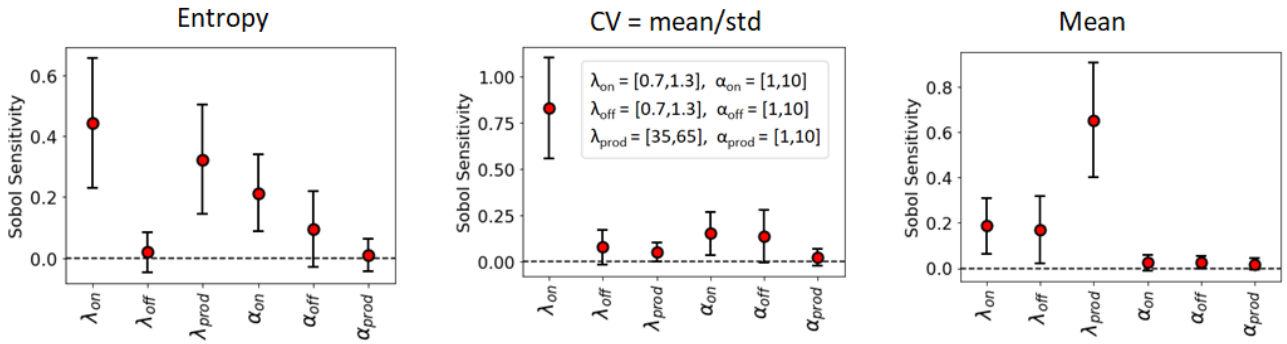


Figure S8: Sobol variance-based sensitivity analysis of the Mean, CV and entropy of the steady state single cell mRNA distribution generated by the two step model, with a grid of 128 points per parameters. The parameter bounds used for all analysis are given in the middle box.

distributions:

$$H(X) = -\int_{\mathcal{X}} f(x)\ \log f(x)\ dx. \qquad [25]$$

Numerically, we compute it by discretizing the mRNA distribution into N_b bins and computing the histogram. As the choice of the number of bin can affect the measure, we chose the same number of bins $N_b = 25$ for all entropy estimations.

The sensitivity analysis of the mean steady state mRNA distribution is consistent with previous analytical results showing that, in the steady state, the mean mRNA distribution only depends on the mean inter-event time distribution [10], and thus, is not influenced by the shape parameters. We also compute phase diagrams

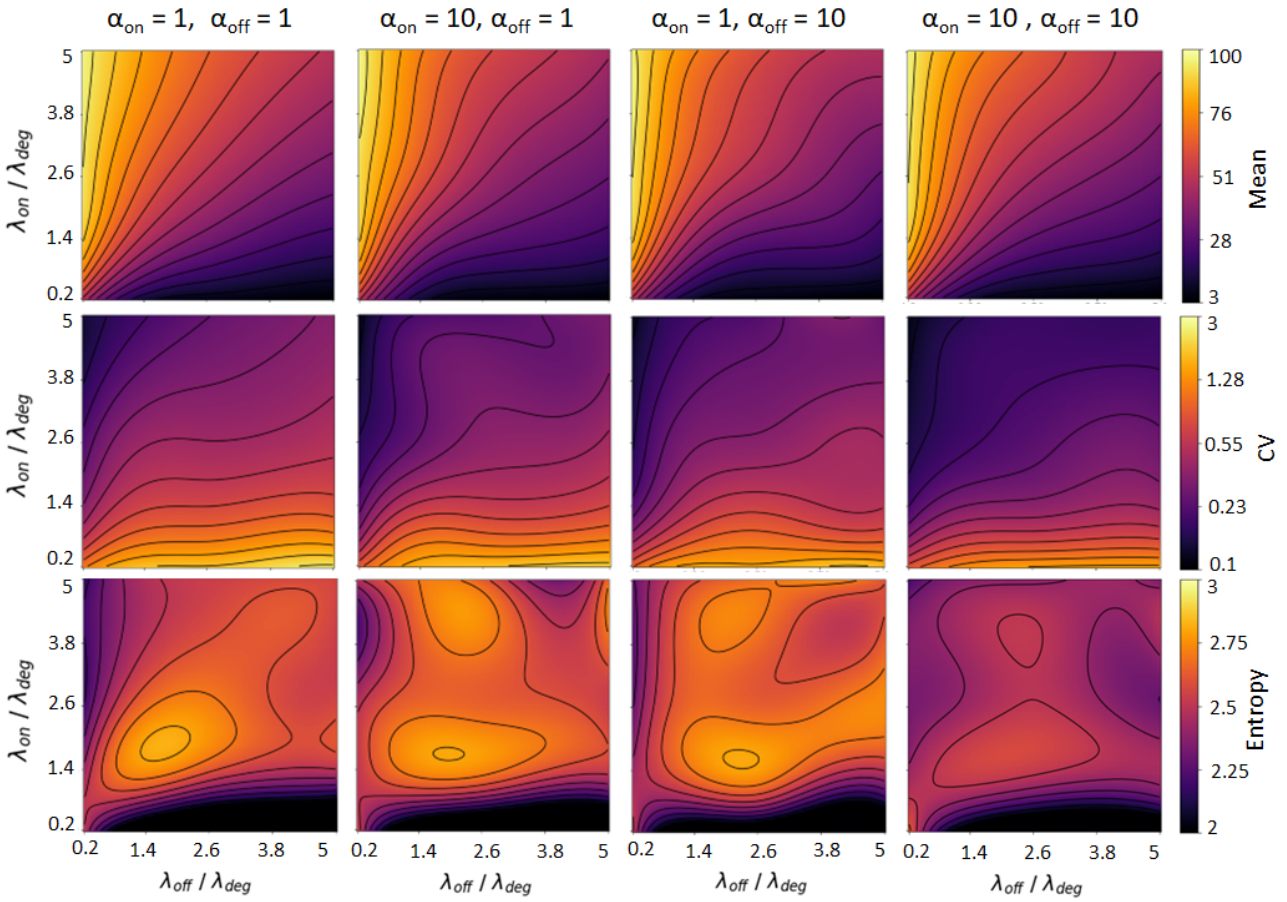


Figure S9: Mean, CV and entropy phase diagram of the steady state single cell mRNA distribution for different combination of shape parameter, i.e. the Markovian ($\alpha_{\text{on}} = \alpha_{\text{off}} = 1$) and non-Markovian cases ($\alpha_{\text{on}} \neq 1, \alpha_{\text{off}} \neq 1$). The heat map were generated with a 2D cubic interpolation of a 10x10 parameter grid, with NoMaSS results averaged over 1000 simulations. Parameters $\alpha_{\text{prod}} = 1$ and $\lambda_{\text{prod}} = 100$ were fixed for all simulations.

of the mean, CV and entropy of mRNA distribution, where the impact of variation in the parameters can be assessed visually. On Figure S9, we show it for the combinations $(\alpha_{\text{on}} = 1, \alpha_{\text{off}} = 1)$, $(\alpha_{\text{on}} = 10, \alpha_{\text{off}} = 1)$, $(\alpha_{\text{on}} = 1, \alpha_{\text{off}} = 10)$ and $(\alpha_{\text{on}} = 10, \alpha_{\text{off}} = 10)$. We observe that, although sensitive, changing the shape parameters (α_{on}, α_{on}) do not affect significantly the phase diagrams of the CV, but rather lower all CV values by a constant factor (0.4 in this case). As for the entropy, the variations across the depicted parameter space is relatively small except for a low λ_{on} where it shows a drastic decrease. We note that the contour plot in the phase diagrams are relatively noisy due to the low variation of the entropy in the simulated parameter range.

Bibliography

[1] John T Sauls et al. “Control of Bacillus subtilis replication initiation during physiological transitions and perturbations”. In: *MBio* 10.6 (2019), e02205–19.

[2] Fangwei Si et al. “Mechanistic origin of cell-size control and homeostasis in bacteria”. In: *Current Biology* 29.11 (2019), pp. 1760–1770.

[3] Yossi Rubner, Carlo Tomasi, and Leonidas J Guibas. "A metric for distributions with applications to image databases". In: *Sixth International Conference on Computer Vision (IEEE Cat. No. 98CH36271)*. IEEE. 1998, pp. 59–66.

[4] Srividya Iyer-Biswas et al. "Scaling laws governing stochastic growth and division of single bacterial cells". In: *Proceedings of the National Academy of Sciences* 111.45 (2014), pp. 15912–15917.

[5] Farshid Jafarpour et al. "Bridging the timescales of single-cell and population dynamics". In: *Physical Review X* 8.2 (2018), p. 021007.

[6] Jennifer C England et al. "Global regulation of gene expression and cell differentiation in Caulobacter crescentus in response to nutrient availability". In: *Journal of bacteriology* 192.3 (2010), pp. 819–833.

[7] Patrick S Stumpf et al. "Stem cell differentiation as a non-Markov stochastic process". In: *Cell Systems* 5.3 (2017), pp. 268–282.

[8] King Davis. "A Global Optimization Algorithm Worth Using". In: *dlib C++ library*. 2017. URL: http://blog.dlib.net/2017/12/a-global-optimization-algorithm-worth.html.

[9] Amit Zeisel et al. "Coupled pre-mRNA and mRNA dynamics unveil operational strategies underlying transcriptional responses to stimuli". In: *Molecular systems biology* 7.1 (2011), p. 529.

[10] Zihao Wang, Zhenquan Zhang, and Tianshou Zhou. "Analytical results for non-markovian models of bursty gene expression". In: *Physical Review E* 101.5 (2020), p. 052406.